

Lecture 3.1

System of Particles and Solid Body

- We've applied Newton's laws to point mass
But, many applications concern
extended objects : cars, rockets,
people...
- cars, rockets, people...=system of
particles or rigid body



**Center of Mass moves in a
straight line. (white dots)**

Other parts rotate around CM

Figure 9.24 Multiflash photograph showing an overhead view of a wrench moving on a horizontal surface. The white dots are located at the center of mass of the wrench and show that the center of mass moves in a straight line as the wrench rotates.

$$\begin{array}{|c|} \hline \text{MOTION of} \\ \text{AN OBJECT} \\ \text{or SYSTEM OF PARTICLES} \\ \hline \end{array} = \begin{array}{|c|} \hline \text{MOTION} \\ \text{of the CENTER of MASS} \\ \hline \end{array} + \begin{array}{|c|} \hline \text{MOTION} \\ \text{of individual particles} \\ \text{Relative to the CENTER of MASS} \\ \hline \end{array}$$

1. Center of Mass (CM)
2. Motion of Center of Mass
3. Conservation of system momentum
4. Kinetic Energy of a System
5. Collisions (elastic, inelastic, perfectly inelastic)
6. Center of Mass Reference frame
7. Motion of Rocket

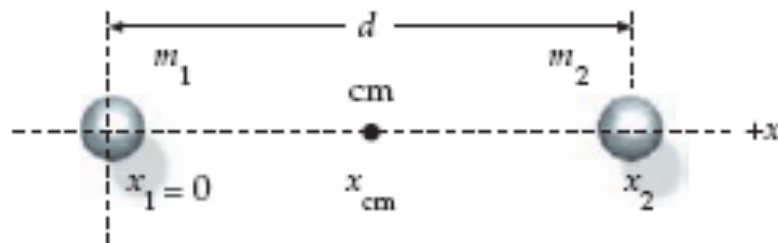
1. THE CENTER OF MASS

System of two particles in one dimension.

Two point masses, m_1 and m_2 , have coordinates x_1 and x_2 on the x axis

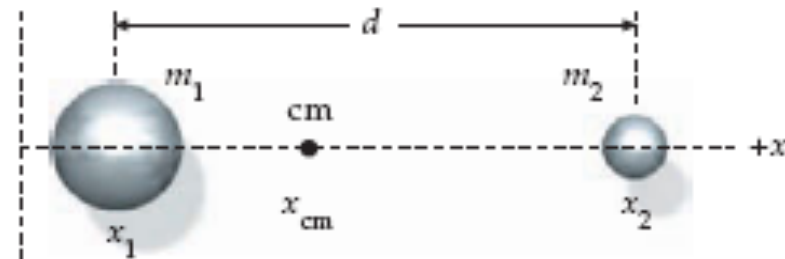
The center-of-mass coordinate x_{cm} is defined by:

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$



(a)

- if the particles have equal masses, the center of mass is midway between them

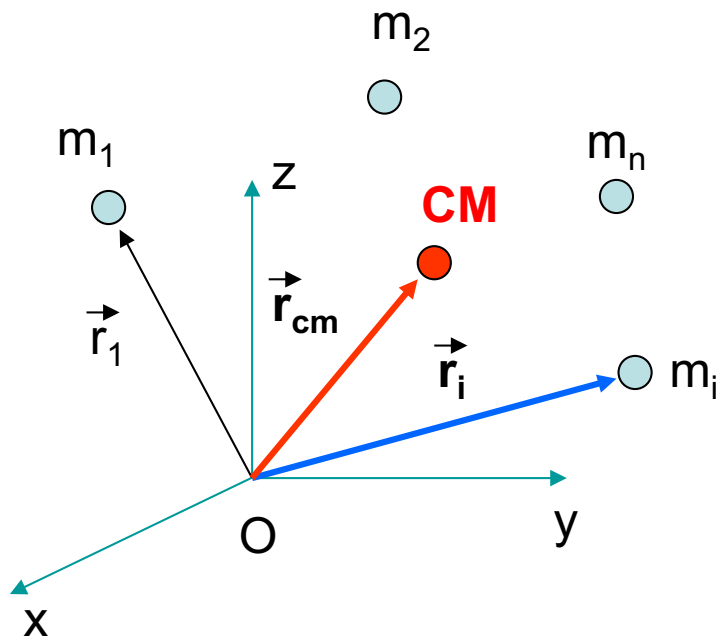


(b)

- If the particles are of unequal mass, the center of mass is closer to the more massive particle

We can generalize from two particles in one dimension to a system of n particles in three dimensions.

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$



$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + \dots + m_i x_i + m_n x_n}{m_1 + m_2 + \dots + m_i + \dots + m_n} = \frac{\sum_i m_i x_i}{\sum_i m_i}$$

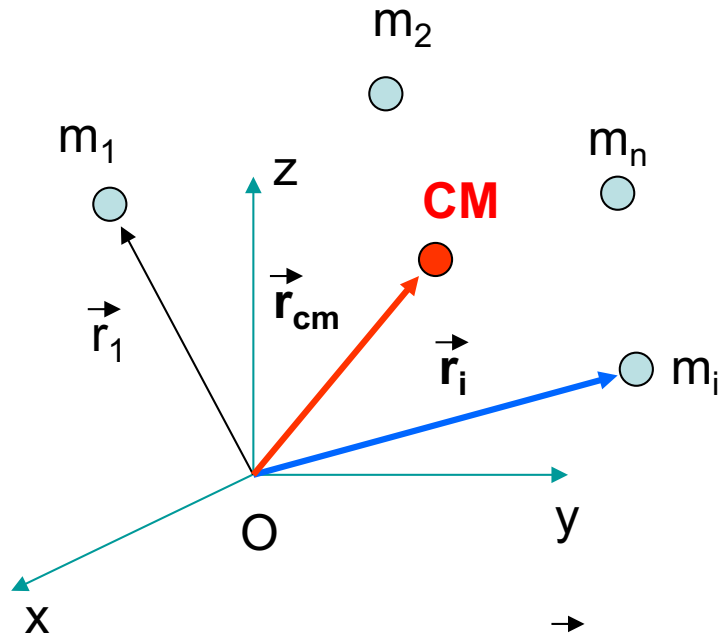
$$y_{cm} = \frac{m_1 y_1 + m_2 y_2 + \dots + m_i y_i + m_n y_n}{m_1 + m_2 + \dots + m_i + \dots + m_n} = \frac{\sum_i m_i y_i}{\sum_i m_i}$$

$$z_{cm} = \frac{m_1 z_1 + m_2 z_2 + \dots + m_i z_i + m_n z_n}{m_1 + m_2 + \dots + m_i + \dots + m_n} = \frac{\sum_i m_i z_i}{\sum_i m_i}$$

$$\vec{r}_i = x_i \vec{i} + y_i \vec{j} + z_i \vec{k}$$

$$\vec{r}_{cm} = x_{cm} \vec{i} + y_{cm} \vec{j} + z_{cm} \vec{k}$$

$$\vec{r}_{cm} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$



$$M\vec{r}_{cm} = \sum_i m_i \vec{r}_i$$

$$M = \sum_i m_i \quad \text{Total mass of the system}$$

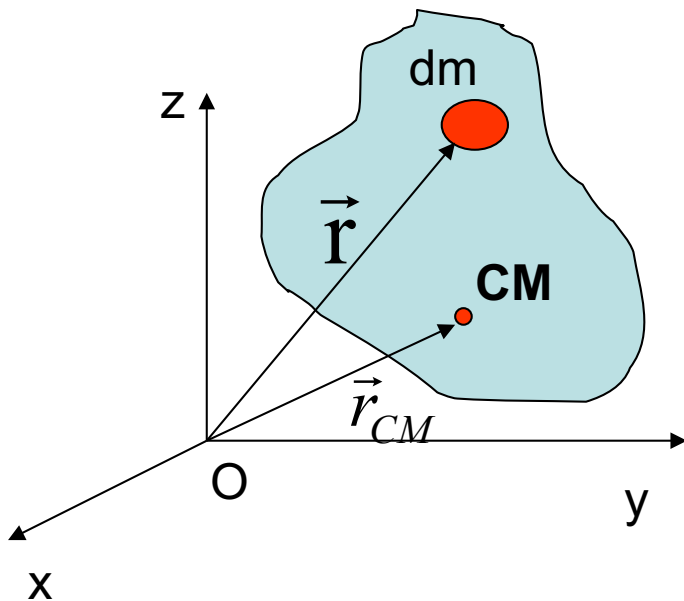
$\vec{r}_i$ is the position vector of the i^{th} particle.

$\vec{r}_{cm}$ is the position vector of the CM.

$$x_{cm} = \frac{\sum_i m_i x_i}{\sum_i m_i} ; \quad y_{cm} = \frac{\sum_i m_i y_i}{\sum_i m_i} ; \quad z_{cm} = \frac{\sum_i m_i z_i}{\sum_i m_i}$$

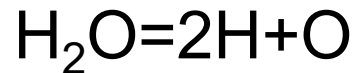
To Find the center of mass of a continuous object

- Divide the object to have an element of mass dm .
- Position vector of dm is $\vec{r}$
- we replace the sum with an integral



$$\vec{r}_{CM} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i} = \frac{\int_{(m)} \vec{r} dm}{\int_{(m)} dm}$$
$$x_{CM} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{\int_{(m)} x dm}{\int_{(m)} dm}; y_{CM} = \frac{\int_{(m)} y dm}{\int_{(m)} dm}; z_{CM} = \frac{\int_{(m)} z dm}{\int_{(m)} dm}$$

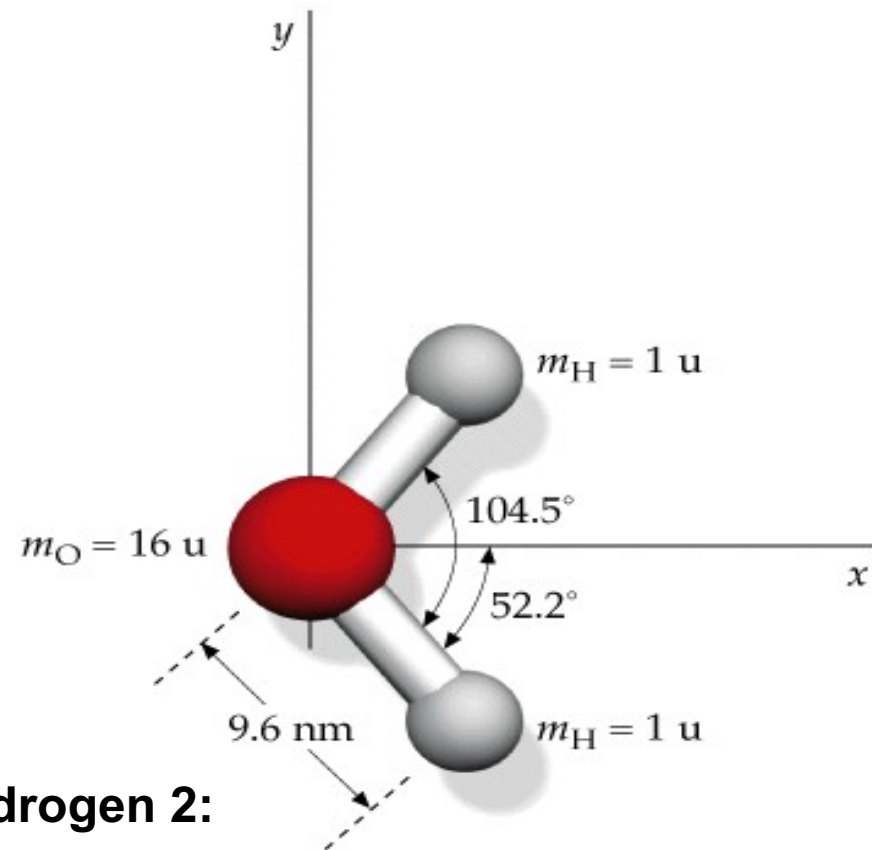
Example 8-1 Find the center of mass of a water molecule $1\text{u}=1.66\times 10^{-27}\text{ kg}$



$$x_{CM} = \frac{\sum_i m_i x_i}{\sum_i m_i} = \frac{m_O x_O + m_{H1} x_{H1} + m_{H2} x_{H2}}{m_O + m_{H1} + m_{H2}}$$

By symmetry, the center of mass is on the x axis

$$\Rightarrow y_{CM} = 0$$



Coordinates of Oxy, Hydrogen 1 and Hydrogen 2:

Oxy (0,0)

H1 ($d\cos 52.2^\circ$; $d\sin 52.2^\circ$)

H2 ($d\cos 52.2^\circ$; $-d\sin 52.2^\circ$);

$d = 9.6\text{nm}$

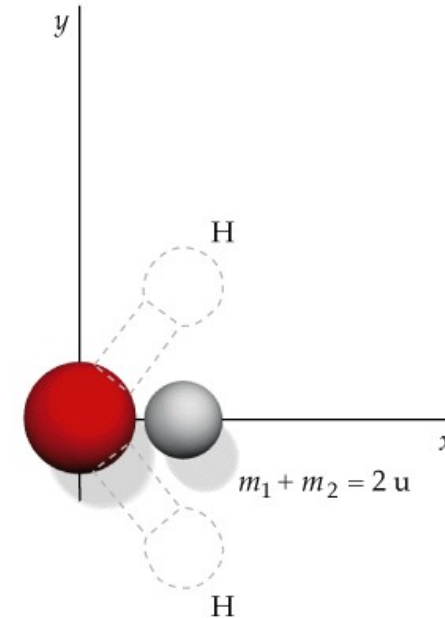
$$x_O = 0$$

$$x_{H1} = x_{H2} = 9.6\text{nm}\cos 52.2^\circ = 5.9\text{nm}$$

$$\begin{aligned} x_{cm} &= \frac{1\text{u} \times 5.9\text{nm} + 1\text{u} \times 5.9\text{nm} + 16\text{u} \times 0}{1\text{u} + 1\text{u} + 16\text{u}} \\ &= 0.66\text{nm} \end{aligned}$$

Another way:

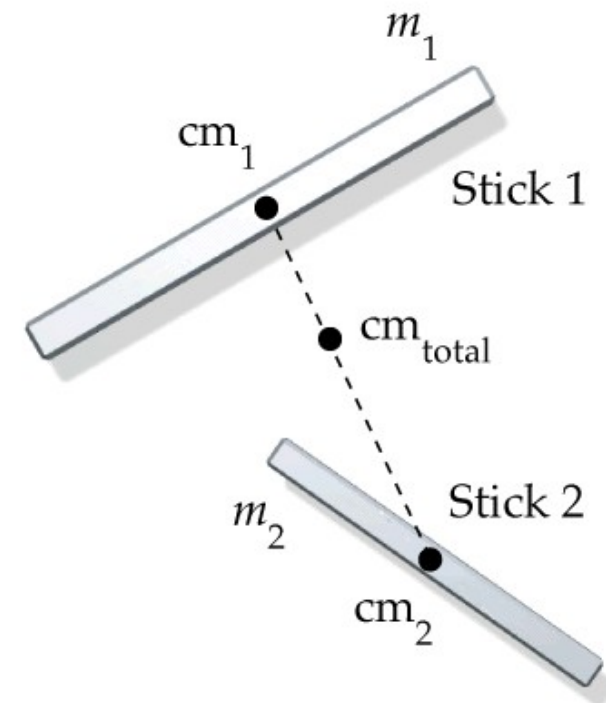
- Finding the center of mass of just the two hydrogen atoms. the two H atoms replaced by a single particle of mass $m_1 + m_2 = 2u$ on the x axis at the center of mass of the original atoms.
- The center of mass then falls between the oxygen atom at the origin and the calculated center of mass of the two hydrogen atoms.



Calculate centers of mass for more complex systems, such as two uniform sticks.

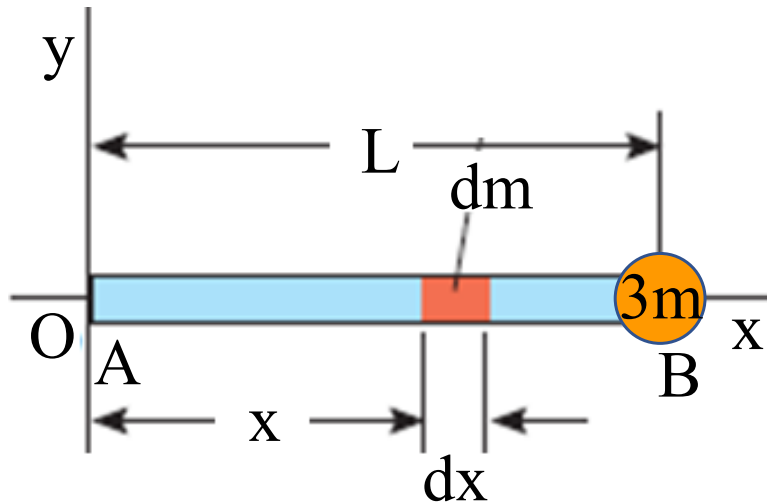
The center of mass of each stick separately is at the center of the stick.

The center of mass of the system is found by treating each stick as a point particle at its individual center of mass.



Example 1

A thin and uniform rod (AB) has a length $L = 80$ cm and mass m uniformly distributed. A small object with a mass of $3m$ is attached to point B of the rod. How far is the center of mass of the system (including the rod and the object) from the point A of the rod?



2. Motion of the Center of Mass

1. The position vector of the center of mass
2. We differentiate (1) to have the velocity of the center of mass
3. We differentiate (2) to have the acceleration of the center of mass
4. According to Newton's second law:
where $\vec{F}_i$ the net force acting on the i th particle
5. Forces acting on a particle fall into two categories:
 - **internal forces** due to interactions with other particles within the system,
 - and **external forces** due to agents outside the system:
6. Substitute (4) into(3):

$$M\vec{r}_{cm} = \sum_i m_i \vec{r}_i \quad (1)$$

$$M \frac{d\vec{r}_{cm}}{dt} = \sum_i m_i \frac{d\vec{r}_i}{dt}$$
$$M\vec{v}_{cm} = \sum_i m_i \vec{v}_i \quad (2)$$

$$M\vec{a}_{cm} = \sum_i m_i \vec{a}_i \quad (3)$$

$$m_i \vec{a}_i = \vec{F}_i$$

$$\vec{F}_i = m_i \vec{a}_i = \vec{F}_{i,int} + \vec{F}_{i,ext} \quad (4)$$

$$M\vec{a}_{cm} = \sum_i \vec{F}_{i,int} + \sum_i \vec{F}_{i,ext} \quad (5)$$

Motion of the Center of Mass (cont.)

7. According to Newton's third law, for each internal force acting on one particle, there is an equal but opposite force acting on another particle. The internal forces thus occur in pairs of equal and opposite forces. **When we sum over all the particles in the system, the internal forces cancel**

$$M\vec{a}_{cm} = \sum_i \vec{F}_{i,int} + \sum_i \vec{F}_{i,ext} \quad (5)$$

$$\sum_i \vec{F}_{i,int} = 0$$

7. Thus, the center of mass of a system moves like a particle of mass $M = \sum m_i$ under the influence of **the net external force** acting on the system.

$$M\vec{a}_{cm} = \sum_i \vec{F}_{i,ext} \quad (6)$$

3. Conservation of Momentum

1. **The momentum of a particle** is defined as the product of its mass and velocity:
2. **The total momentum of a system** of many particles is the sum of the momenta of particles, and equals the total mass M times the velocity of the center of mass:
3. Differentiating this equation with respect to time, we obtain
4. When the net external force acting on a system of particles is zero, the rate of change of the total momentum is zero, and the total momentum of the system remains constant:

$$\vec{p} = m\vec{v}$$

$$\vec{P} = \sum_i \vec{p}_i = \sum_i m_i \vec{v}_i = M\vec{v}_{cm}$$

$$\frac{d\vec{P}}{dt} = M \frac{d\vec{v}_{cm}}{dt} = M\vec{a}_{cm}$$

$$\frac{d\vec{P}}{dt} = \vec{F}_{net_ext}$$

$$\vec{F}_{net_ext} = 0$$

$$\frac{d\vec{P}}{dt} = 0 \Rightarrow \vec{P} = \overrightarrow{const}$$

5. Collision

- In a collision, two objects approach and interact strongly for a very short time.
- During the brief time of collision, **any external forces are much smaller than the forces of interaction between the objects.**

$$\vec{F}_{\text{ext_net}} = 0$$

- **Momentum of the system is conserved**

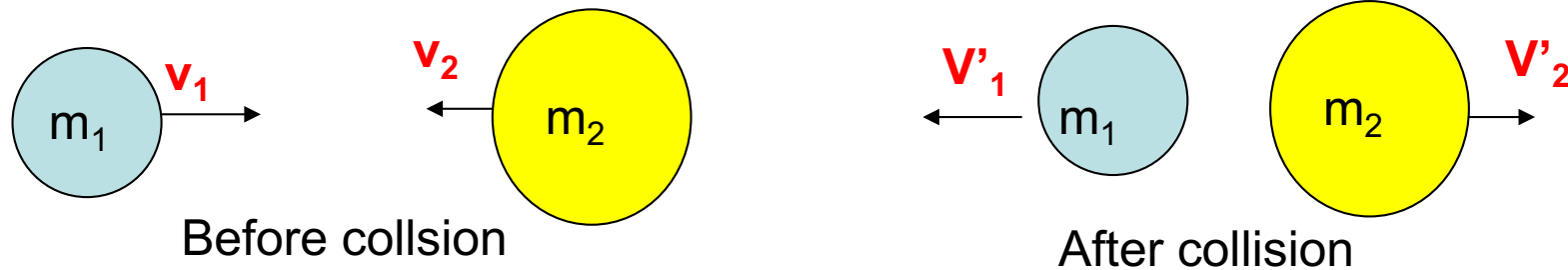
$$\frac{d\vec{P}}{dt} = \vec{F}_{\text{net_ext}} = 0 \Rightarrow \vec{P} = \overrightarrow{\text{const}}$$

Watch this video: <https://www.youtube.com/watch?v=M2xnGcaaAi4>

Classification:

- elastic collision (total kinetic energy is conserved)
- Inelastic collision (total kinetic energy is not conserved)
- perfectly inelastic collisions (after collision, 2 objects stick together and have the same velocity)

6a. Elastic collision



- **Conservation of momentum:**

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = m_1 \vec{v}'_1 + m_2 \vec{v}'_2 \quad (1)$$

$$m_1 (\vec{v}_1 - \vec{v}'_1) = m_2 (\vec{v}'_2 - \vec{v}_2) \quad (2)$$

- **For elastic collisions, conservation of kinetic energies**

$$\frac{1}{2} m_1 \vec{v}_1^2 + \frac{1}{2} m_2 \vec{v}_2^2 = \frac{1}{2} m_1 \vec{v}'_1^2 + \frac{1}{2} m_2 \vec{v}'_2^2 \quad (3)$$

$$m_1 (\vec{v}_1^2 - \vec{v}'_1^2) = m_2 (\vec{v}'_2^2 - \vec{v}_2^2) \quad (4)$$

- **Eq(6) The relative velocity of two particles before and after collision has the same magnitude**

$$\vec{v}_1 + \vec{v}'_1 = \vec{v}'_2 + \vec{v}_2 \quad (5)$$

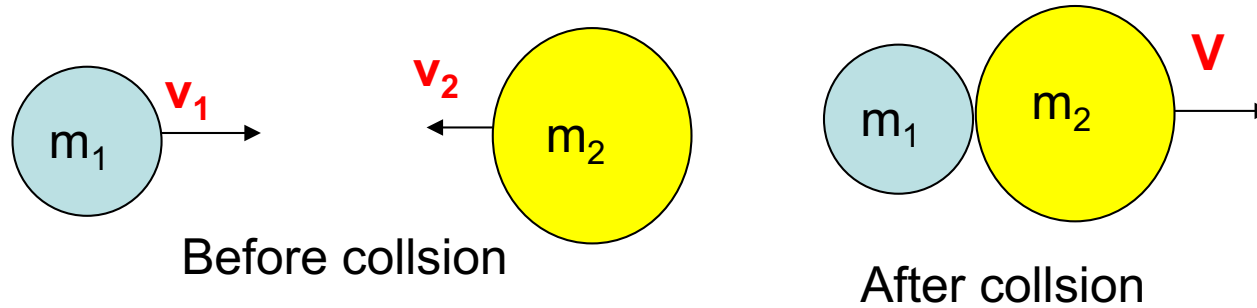
$$(\vec{v}_1 - \vec{v}_2) = -(\vec{v}'_1 - \vec{v}'_2) \quad (6)$$

- **From(2) and (6) , the velocity of the two particles after collision**

$$\vec{v}'_1 = \frac{(m_1 - m_2) \vec{v}_1 + 2m_2 \vec{v}_2}{m_1 + m_2}$$

$$\vec{v}'_2 = \frac{(m_2 - m_1) \vec{v}_2 + 2m_1 \vec{v}_1}{m_1 + m_2}$$

6b. Perfectly Inelastic Collisions in One Dimension



- **Conservation of momentum:**

$$m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{V} \quad (1)$$

- **After collision, two objects stick together, and has the same velocity**

$$\vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \quad (2)$$

Lecture 3.2

ROTATION

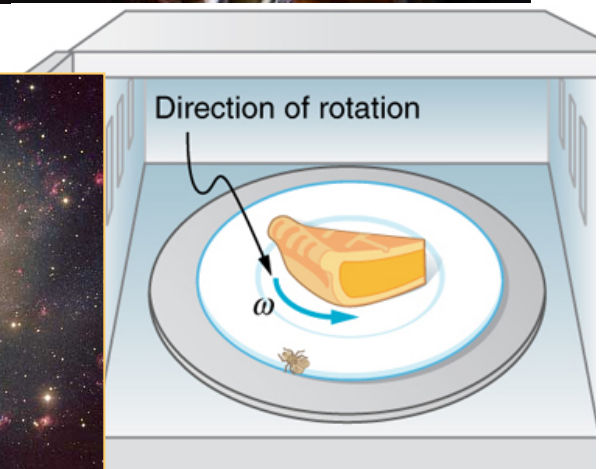
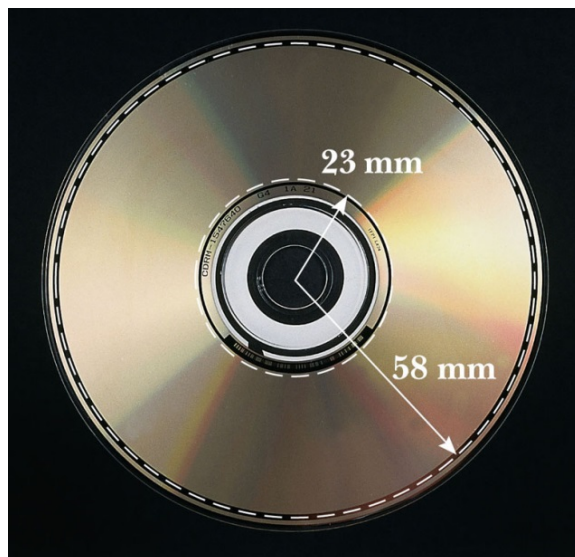
<https://www.youtube.com/watch?v=grMWAI1RdVs>

Outline

- Cross product
- Rotation about a fixed axis
- Angular Velocity-Angular Acceleration
- Torque
- Equation of Rotation
- Moment of Inertia
- Kinetics Energy of Rotation

Rotation in Reality

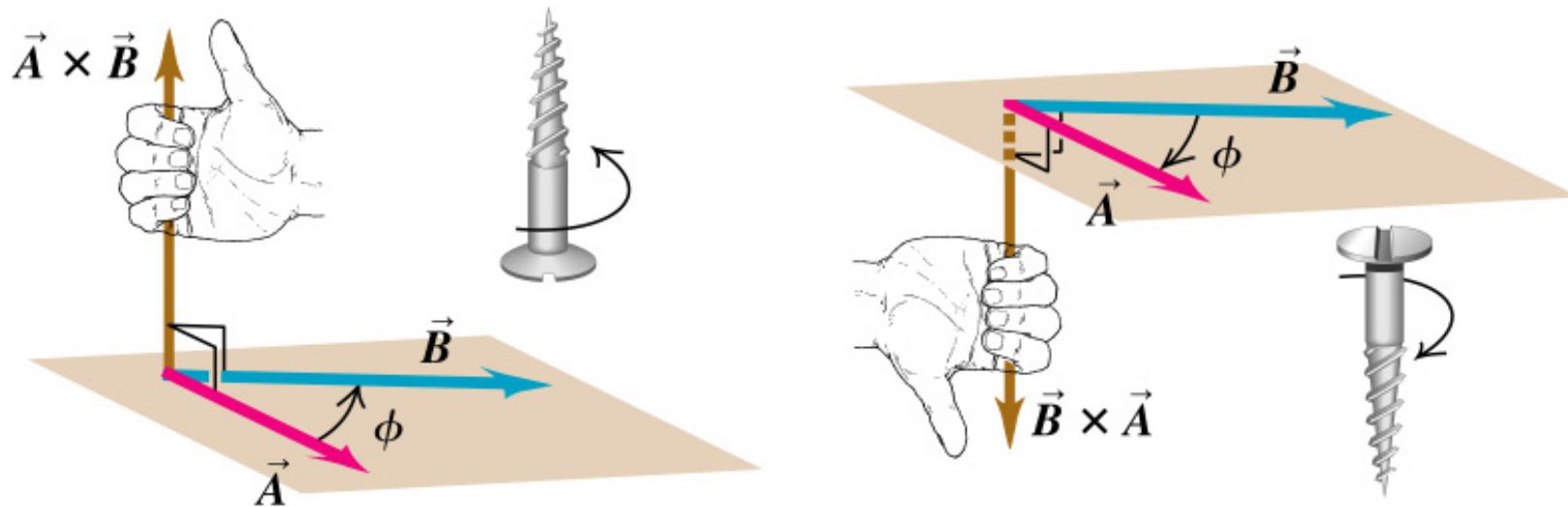
- Rotational motion is all around us from molecules to galaxies.
- The earth rotates about its axis.
- Wheels, gears (sự truyền động bánh răng)
- Propellers (cánh quạt máy bay),
- Motors, a CD in its player, a pirouetting (múa xoay tròn) ice skater, all rotate.



Cross Product

$$\vec{C} = \vec{A} \times \vec{B}$$

$$C = AB \sin \phi$$



Properties of the Cross Product

- Cross Product is **Anti-Commutative**.
- **Parallel** Vectors have Cross Product of **zero**.
- Cross Product obeys the **Distributive Law**.
- **Product Rule** for **Derivative** of a Cross Product.

$$\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$$

$$\frac{d}{dt}(\vec{A} \times \vec{B}) = \vec{A} \times \frac{d\vec{B}}{dt} + \frac{d\vec{A}}{dt} \times \vec{B}$$

Cross Product

- The **Cross Product** (or **Vector Product**) of two vectors **A** and **B** is a multiplication of vectors where the result is a **vector** quantity **C** with a direction **perpendicular** to both vectors **A** and **B**, and the magnitude equal to $AB\sin\phi$:

$$\vec{C} = \vec{A} \times \vec{B}$$

$$C = AB \sin \phi$$
 Magnitude of the cross (vector) product of two vectors **A** and **B**

- A is the magnitude of the first vector, B is the magnitude of the second vector and ϕ is the angle between the two vectors.
- The direction of the cross product is perpendicular to the plane formed by the two vectors in the product. This leaves two possible choices which are resolved by using the **Right Hand Rule**.

Cross Product

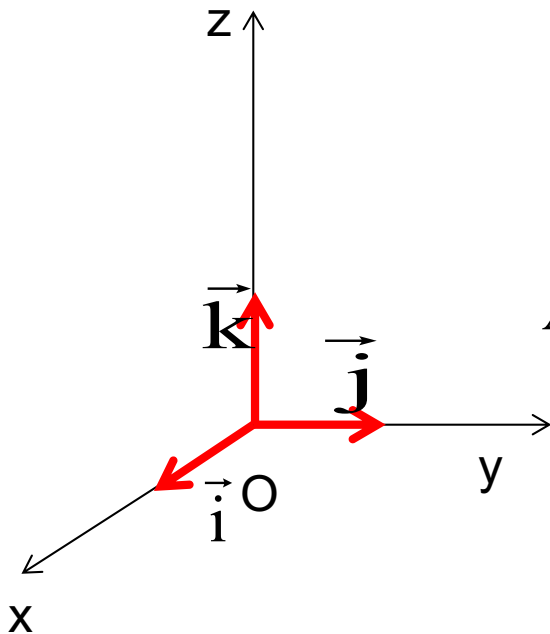
- **Right-handed** coordinate system, in which:

$$\vec{i} \times \vec{j} = \vec{k}$$

$$\vec{j} \times \vec{k} = \vec{i}$$

$$\vec{k} \times \vec{i} = \vec{j}$$

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k}$$



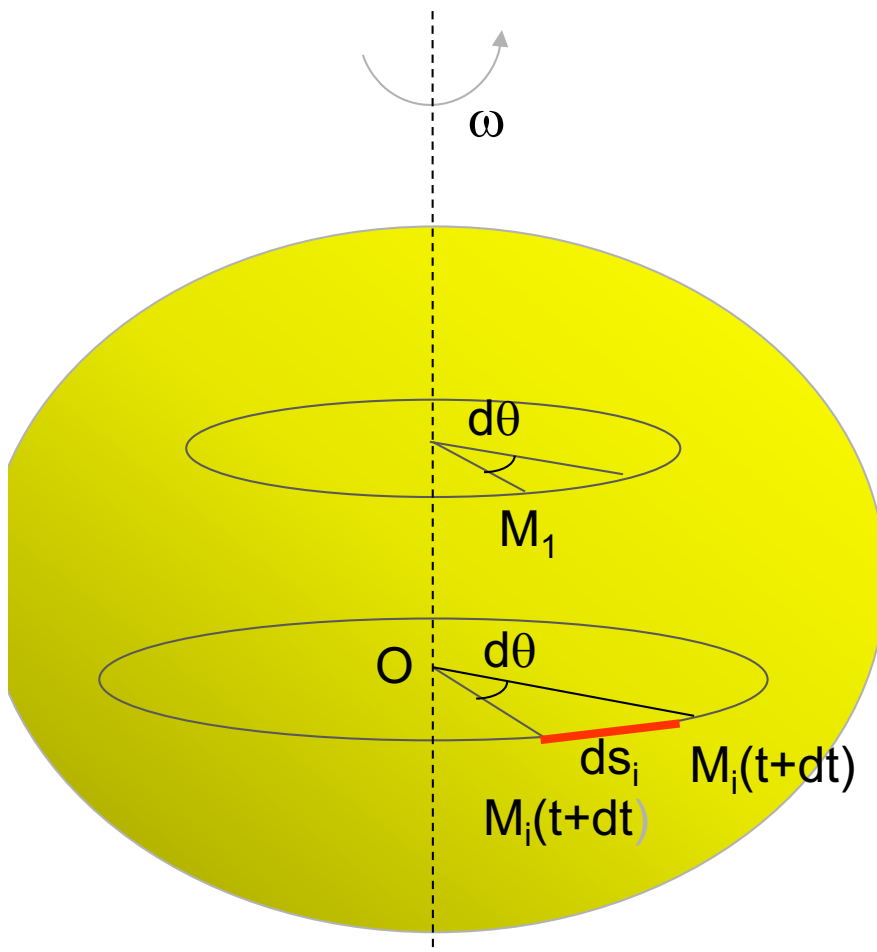
ANGULAR VELOCITY - ANGULAR ACCELERATION

A rigid body = system of particles, that the distance between any two particles that make up the object remains fixed.

Rigid object rotates about a fixed axis Δ .

The trajectory of each point (except those on the axis) is a circle.

During a time dt , all points move the same angle $d\theta$



Angular speed

$$\omega = \frac{d\theta}{dt} (\text{rad} / \text{s})$$

Angular
Acceleration

$$\alpha = \frac{d\omega}{dt} (\text{rad} / \text{s}^2)$$

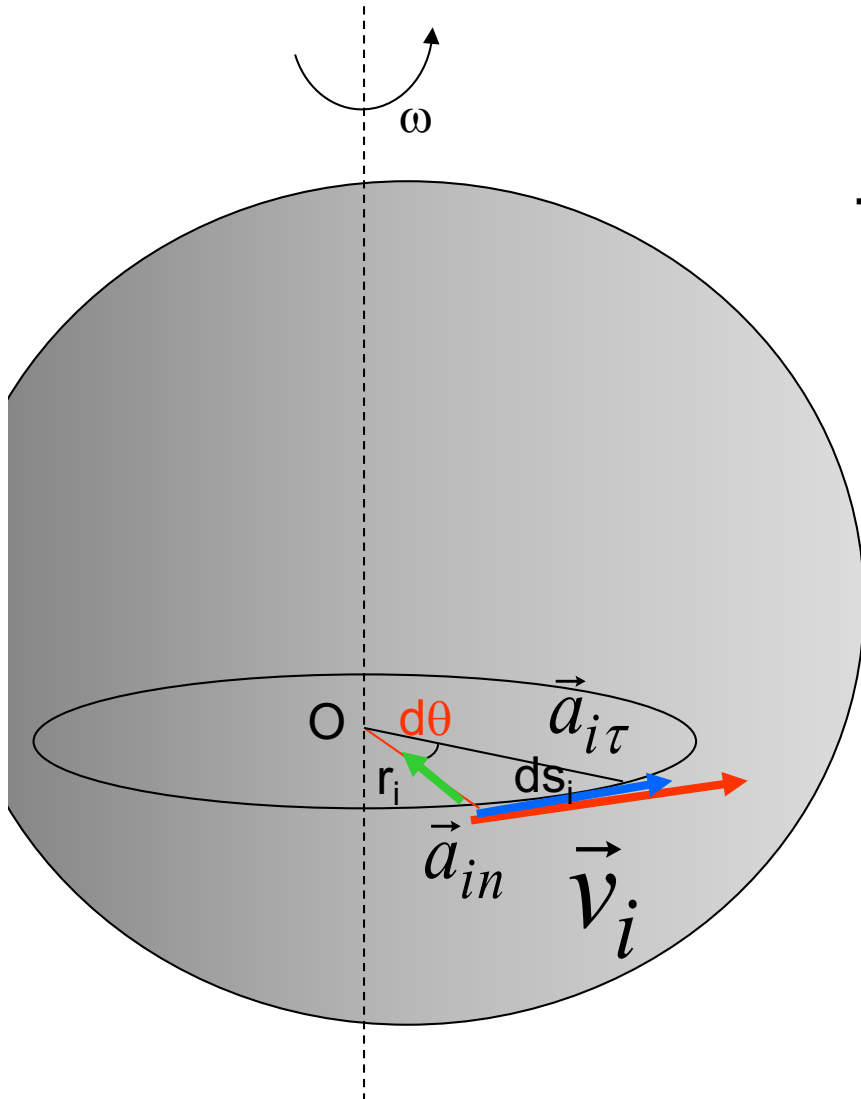
During a time dt , i^{th} particle move a distance ds_i

$$ds_i = r_i d\theta$$

r_i : distance from i^{th} particle M_i to the rotation axis.

Relationship Between Angular and Linear Quantities

$$ds_i = r_i d\theta$$



The speed of the i^{th} particle:

$$v_i = \frac{ds_i}{dt} = r_i \frac{d\theta}{dt} = r_i \omega$$

The tangential acceleration of the i^{th} particle :

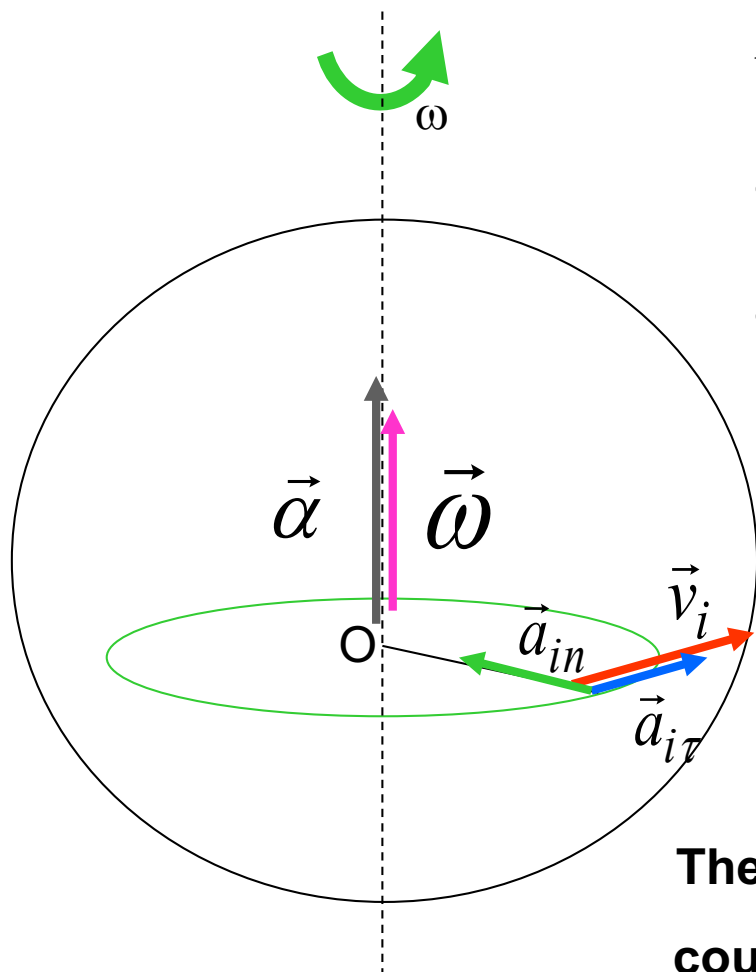
$$a_{ti} = \frac{dv_i}{dt} = \frac{d\omega}{dt} r_i = \alpha r_i$$

The centripetal acceleration of the i^{th} particle :

$$a_{in} = \frac{v_i^2}{r_i} = \omega^2 r_i$$

The acceleration of the i^{th} particle

$$a_i = \sqrt{a_{ti}^2 + a_{ni}^2} = r_i \sqrt{\alpha^2 + \omega^4}$$



$$\vec{v}_i = \vec{\omega} \wedge \vec{r}_i$$

$$\vec{a}_{i\tau} = \vec{\alpha} \wedge \vec{r}_i$$

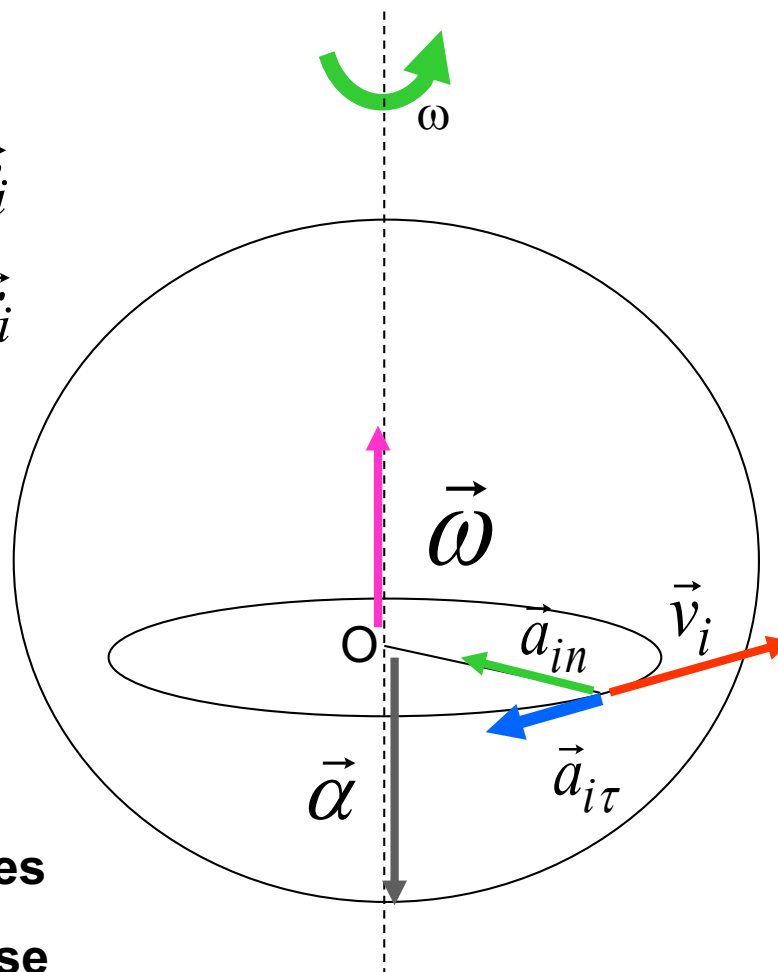
$$\vec{a}_{in} = -\omega^2 \vec{r}_i$$

The object rotates
counter-clockwise

$$\vec{\alpha} \uparrow \uparrow \vec{\omega}$$

$$\vec{a}_{i\tau} \uparrow \uparrow \vec{v}_i$$

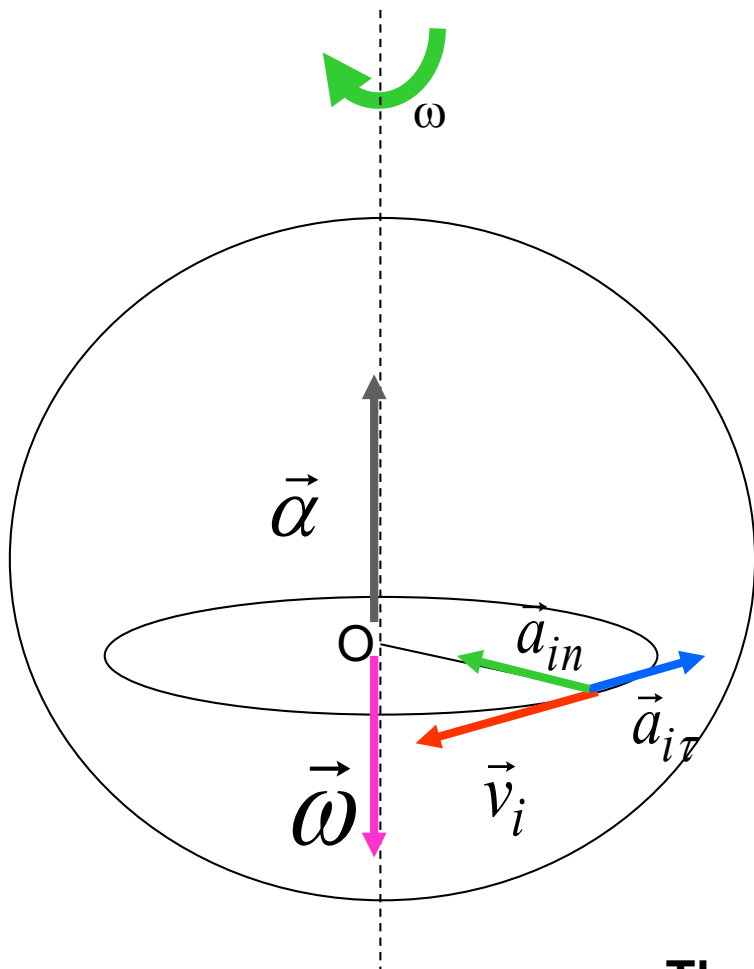
The rotation is accelerated



$$\vec{\alpha} \uparrow \downarrow \vec{\omega}$$

$$\vec{a}_{i\tau} \uparrow \downarrow \vec{v}_i$$

The rotation is decelerated



$\vec{\alpha} \uparrow \downarrow \vec{\omega}$

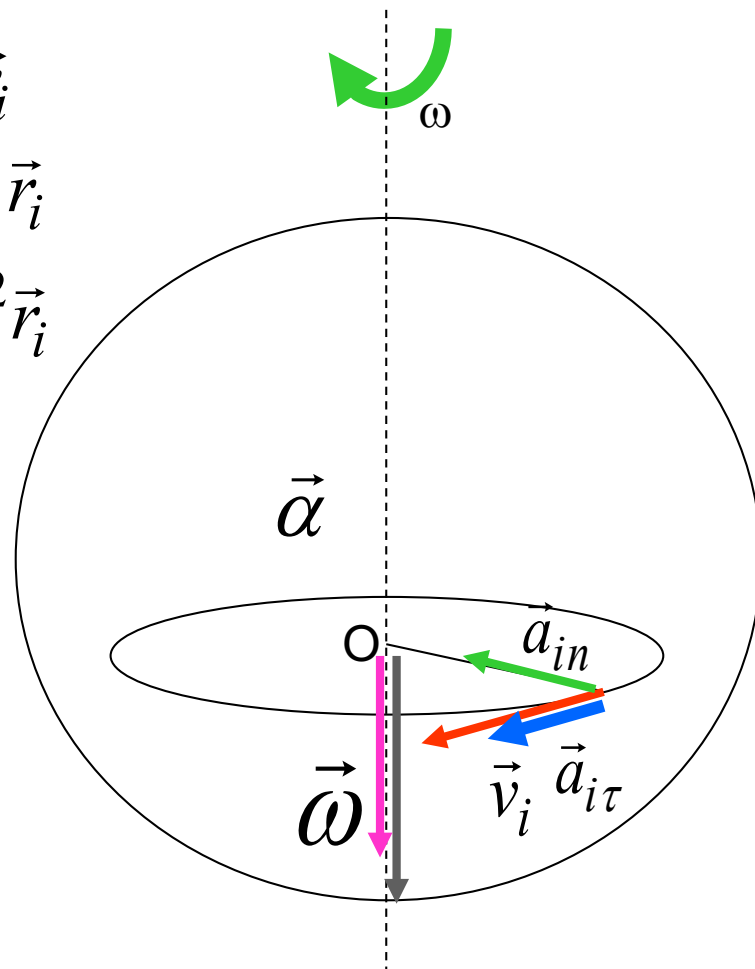
$\vec{a}_{i\tau} \uparrow \downarrow \vec{v}_i$

The rotation is decelerated

$$\vec{v}_i = \vec{\omega} \wedge \vec{r}_i$$

$$\vec{a}_{i\tau} = \vec{\alpha} \wedge \vec{r}_i$$

$$\vec{a}_{in} = -\omega^2 \vec{r}_i$$

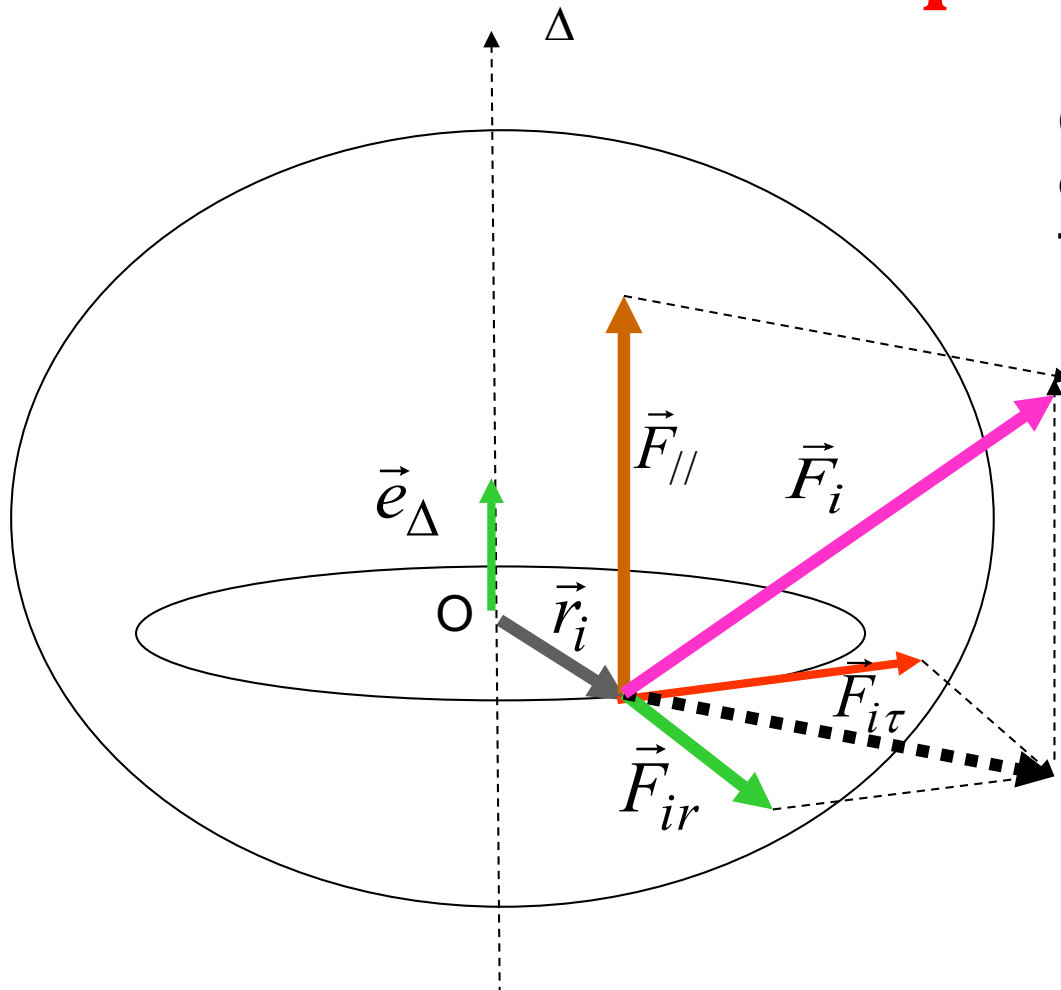


$\vec{\alpha} \uparrow \uparrow \vec{\omega}$

$\vec{a}_{i\tau} \uparrow \uparrow \vec{v}_i$

The rotation is accelerated

Torque



$\vec{e}_\Delta$: unit _ evector

The torque of the Force F about the Δ axis going through the point O

Consider a force $\vec{F}_i$ exerting on the object, that can rotate about a fixed axis.

$$\vec{F} = \vec{F}_{//} + \vec{F}_r + \vec{F}_t$$

Only the tangential force can rotate the object

The torque of the Force F about a fixed point O:

$$\vec{\tau}_{\vec{F}/O} = \vec{r} \wedge \vec{F}$$

$$\vec{\tau}_{\vec{F}/\Delta} = (\vec{\tau}_{\vec{F}/O} \cdot \vec{e}_\Delta) \cdot \vec{e}_\Delta$$

$$\vec{\tau}_{\vec{F}/O} = \vec{r} \wedge \vec{F} = \vec{r} \wedge (\vec{F}_{//} + \vec{F}_r + \vec{F}_t)$$

$$\vec{\tau}_{\vec{F}/O} = \underbrace{(\vec{r} \wedge \vec{F}_{//})}_{-rF_{//}\vec{e}_\tau} + \underbrace{(\vec{r} \wedge \vec{F}_r)}_{=0} + \underbrace{(\vec{r} \wedge \vec{F}_t)}_{rF_t\vec{e}_\Delta}$$

$$\vec{\tau}_{\vec{F}/\Delta} = (\vec{M}_{\vec{F}/O} \cdot \vec{e}_\Delta) \vec{e}_\Delta = rF_t \vec{e}_\Delta = \vec{M}_{\vec{F}_t/\Delta}$$

Only the tangential force can rotate the object

$$\vec{\tau}_{\vec{F}/\Delta} = \vec{\tau}_{\vec{F}_t/\Delta}$$

The torque of a parallel force $F_{//}$ is zero

$$\vec{\tau}_{\vec{F}_{//}/\Delta} = 0$$

The torque of the force that has the line of action go through the axis is zero

$$\vec{\tau}_{\vec{F}_r/\Delta} = 0$$

DYNAMICS OF ROTATION

Let Force F_i exerts on the i th particle

Force F_{it} causes a tangential acceleration a_{it} of the i th particle:

$$F_{it} = m_i a_{it} = m_i r_i \alpha \quad (1)$$

$$a_{it} = r_i \alpha$$

Multiply the eq(1). by r_i

$$r_i F_{it} = m_i r_i^2 \alpha \quad (2)$$

Torque of F_i about the rotation axis

$$\tau_i = m_i r_i^2 \alpha \quad (2')$$

Summing over all the particles in the object

$$\sum_i \tau_i = \left(\sum_i m_i r_i^2 \right) \alpha \quad (3)$$

(Angular acceleration is the same for all the particles of the object and can therefore be taken out of the sum)

$$\tau = \sum_i \tau_i \quad I = \sum_i m_i r_i^2$$

τ : the net torque acting on the object:

I : Moment of Inertia

$$\tau = I \alpha$$

Equation of Rotation

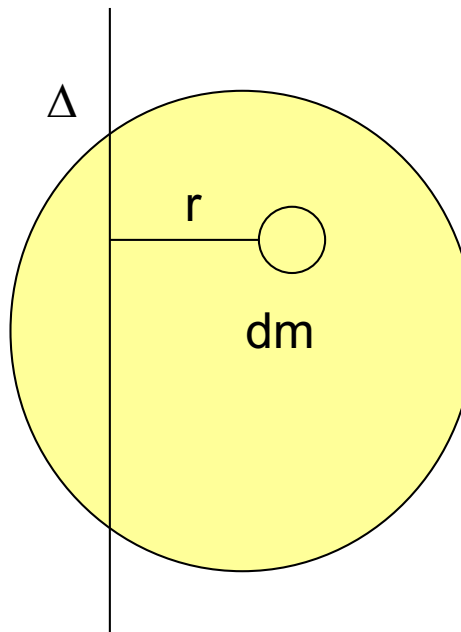
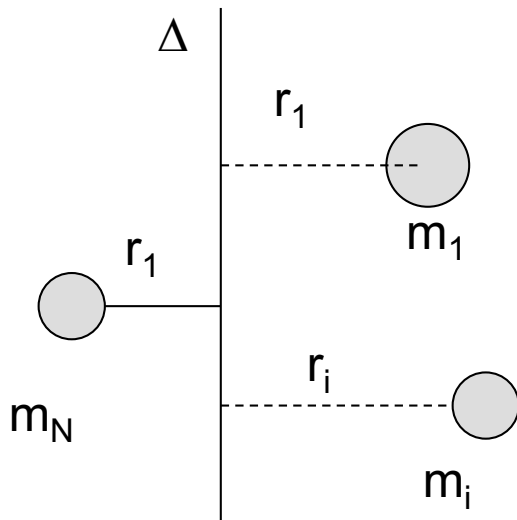
MOMENT OF INERTIA

$$I = \sum_i m_i r_i^2 \quad (\text{system of particles})$$

$$I = \int r^2 dm \quad (\text{continuous object})$$

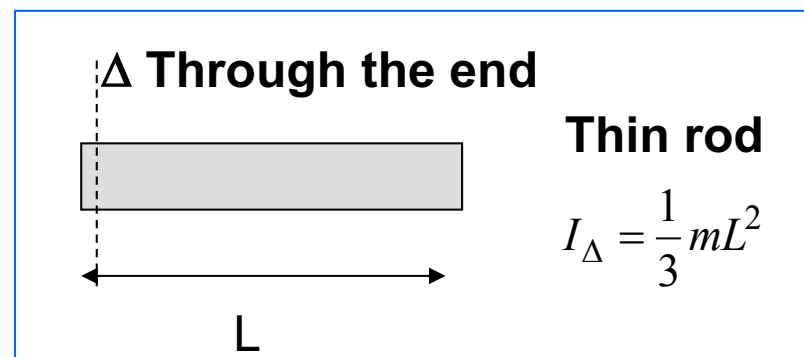
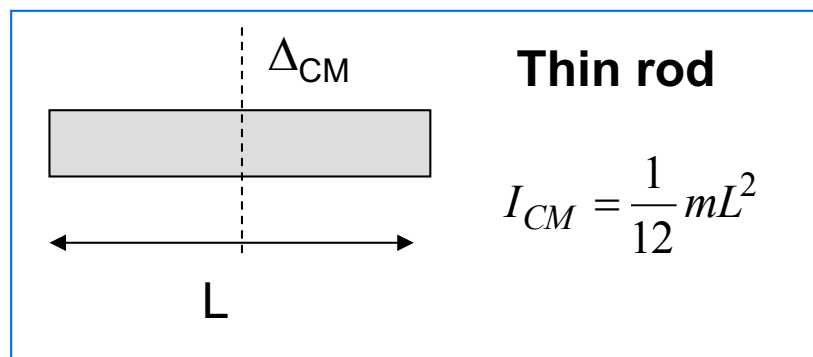
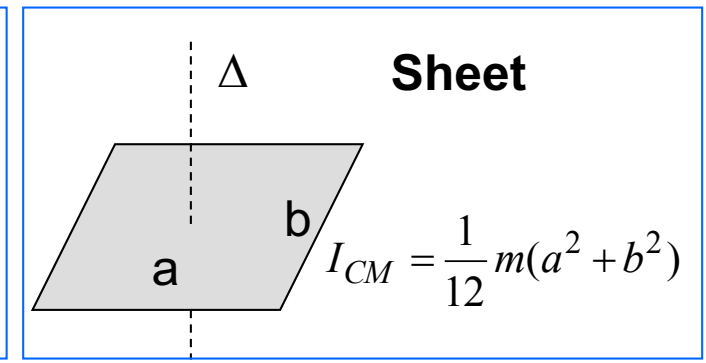
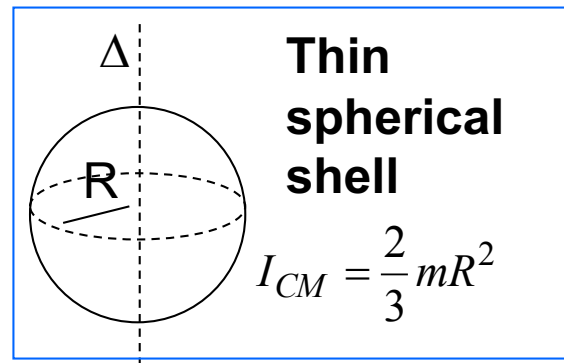
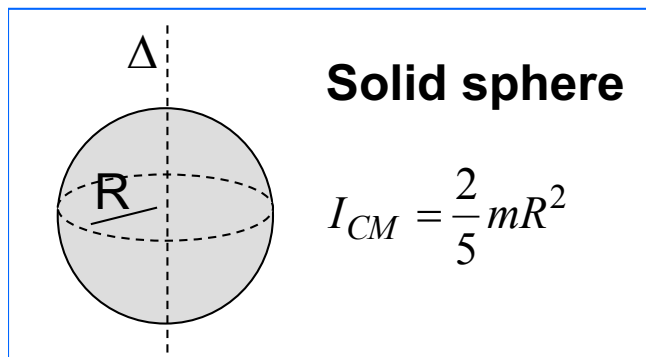
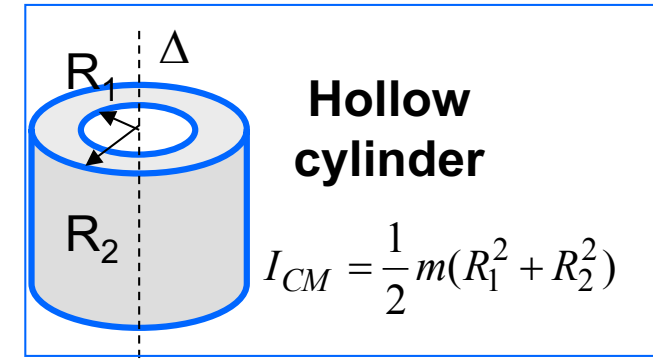
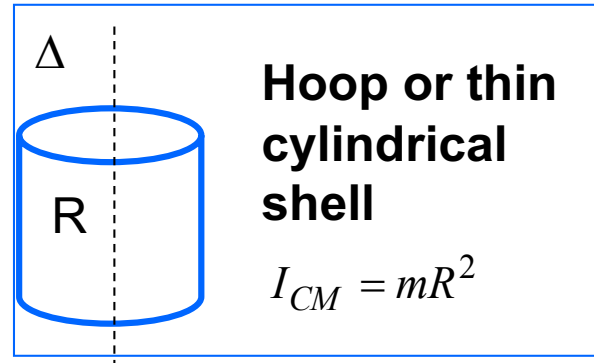
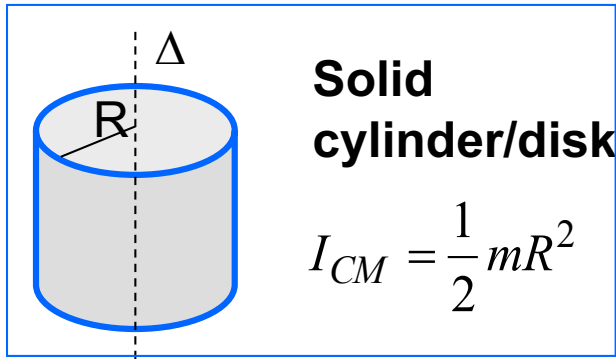
r_i is the distance of the i th particle from the rotation axis Δ

r is the distance of the mass element dm from the rotation axis Δ



Moment of Inertia

of Homogenous Rigid Objects about the axis through the Center of Mass

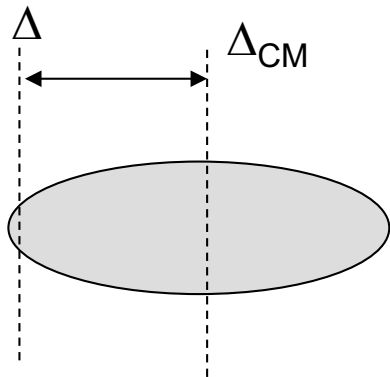


The Parallel-Axis Theorem (Steiner- Huygens)

The moment of inertia about an axis is equal to the sum of moment of inertia about an axis through the center of mass I_{cm} and Md^2

$$I_{\Delta} = I_{CM} + md^2$$

d : distance between through parallel axes Δ and Δ_{cm}



$$I_{\Delta} = I_{CM} + md^2$$

$$I_{CM} = \frac{1}{2}mR^2, d = R$$

$$I_{\Delta} = \frac{1}{2}mR^2 + mR^2 = \frac{3}{2}mR^2$$

Kinetic Energy of Rotation

The kinetic energy of a rotating object is the sum of the kinetic energies of the individual particle in the object.

The kinetic energy of a mass element m_i is

$$K_i = \frac{1}{2} m_i v_i^2$$

Summing over all the elements and using $v_i = r_i \omega$ gives

$$K_{rot} = \sum_i K_i = \sum_i \frac{1}{2} m_i v_i^2 = \sum_i \frac{1}{2} m_i r_i^2 \omega^2 = \frac{1}{2} \left(\sum_i m_i r_i^2 \right) \omega^2$$

$$K_{rot} = \frac{1}{2} I \omega^2$$

Kinetic Energy of Rotation

Work done by the force

$$dW =$$

$$\vec{F} \cdot d\vec{s} = \vec{F} \cdot (d\vec{\theta} \times \vec{r}) = (\vec{r} \times \vec{F}) \cdot d\vec{\theta}$$

Work in rotational motion:

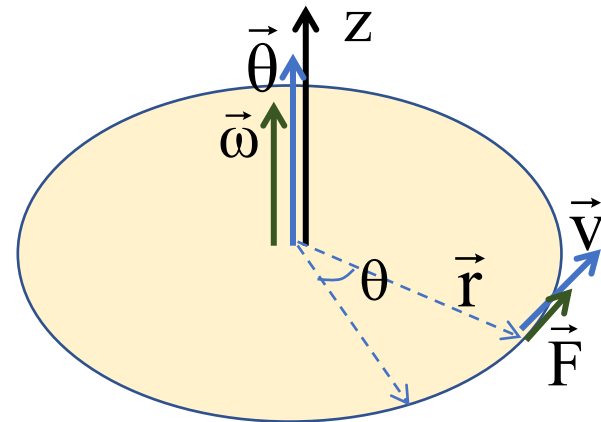
$$W = \int \vec{F} \cdot d\vec{s} = \int \vec{\tau} \cdot d\vec{\theta}$$

$$\text{If } \vec{\tau} = \text{const}, W = \vec{\tau} \cdot \vec{\theta}$$

$$\text{Power: } P = \vec{\tau} \cdot \vec{\omega}$$

Kinetic and work relationship:

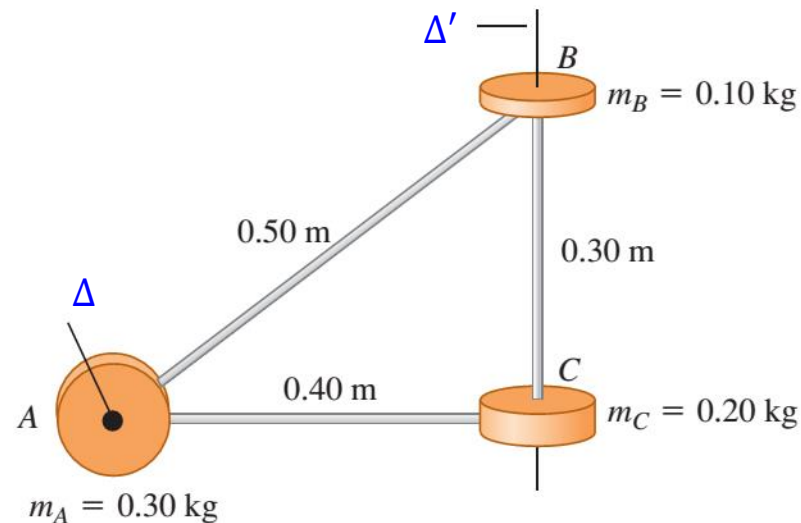
$$W = K_2 - K_1 = \frac{1}{2} I \omega_2^2 - \frac{1}{2} I \omega_1^2$$



Example 2

Given the system as shown on the figure. Consider A, B, C as points. Calculate:

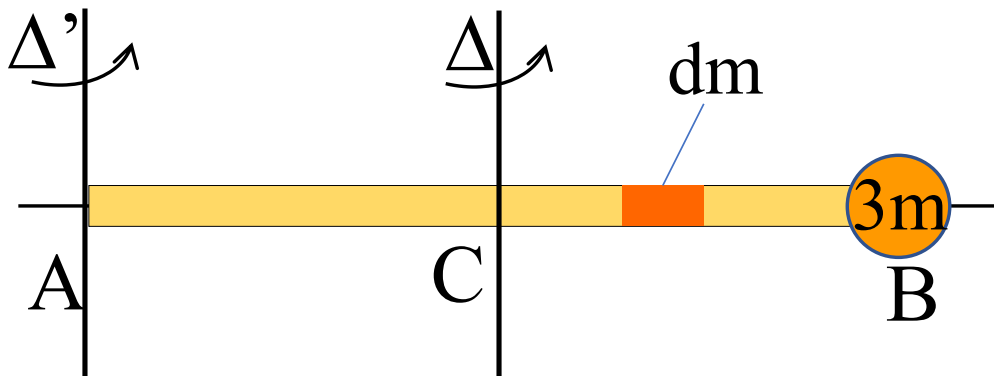
- The moment of inertia of the system about the rotation axis Δ passes through A, which is perpendicular to the plane of the figure.
- The moment of inertia of the system about the rotation axis Δ' passes through B and C.
- Calculate the kinetic energy of the system about the rotating axis through A and having angular velocity $\omega = 4 \text{ rad/s}$.



Example 3

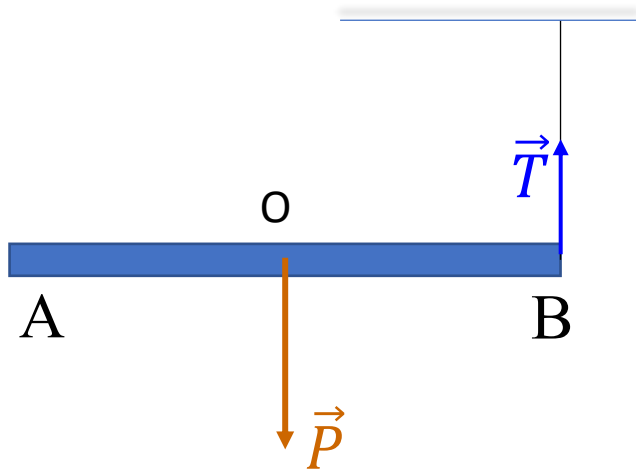
A thin and uniform rod (AB) has a length $L = 80$ cm and mass m uniformly distributed. A small object with a mass of $3m$ is attached to point B of the rod. Calculate the moment of inertia of the system (rod + object) about the axis of rotation

- Δ passes through the center of mass C of the rod and is perpendicular to the rod.
- Δ' passes through point A of the rod and is perpendicular to the rod.



Example 4

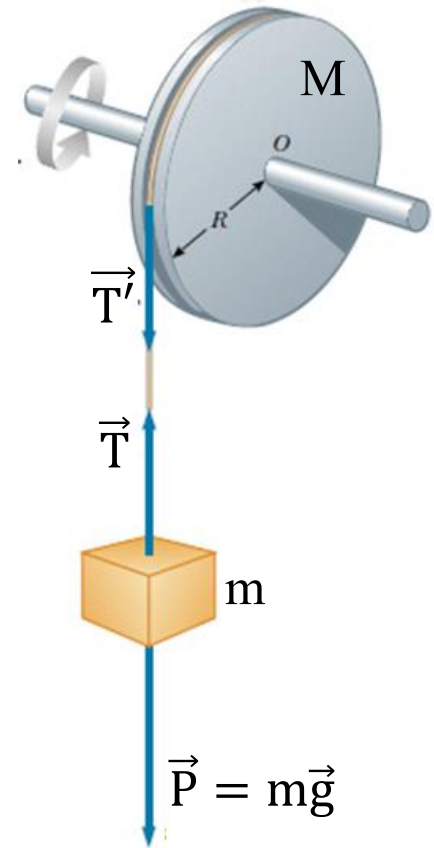
A uniform rod (AB) has a mass of $m = 2 \text{ kg}$, evenly distributed. The rod can rotate around a fixed horizontal axis passing through the end point A, which is perpendicular to the rod. The end point B of the rod is hung by a very light, non-stretchable string. Ignore the friction at the rotation axis, take $g = 10 \text{ m/s}^2$. When the rod is in horizontal equilibrium, the string hangs vertically, what is the tension in the string?



Example 5

A thin, light thread is wound around a uniform solid cylinder of mass M and radius R . A small object of mass m is tied to the end of the thread and released gently to set the system in motion, see the figure. Ignore friction at the rotating axis of the cylinder. Find

- Acceleration of the object m .
- Kinetic energy of the system "cylinder + object" as a function of time.



Example 6

When disconnected from the power source, the fan begins to rotate slowly until it stops. The work and torque of the resisting force are -47.1 J and -0.1 N.m, respectively. What is the number of revolutions that the fan makes from the time the power is turned off until it stops?

Example 1

Two blocks are connected by a string that passes over a pulley of radius R and moment of inertia I . The block of mass m_1 slides on a frictionless, horizontal surface; the block of mass m_2 is suspended from the string. Find the acceleration a of the blocks and the tensions T_1 and T_2 assuming that the string does not slip on the pulley.

Example 2

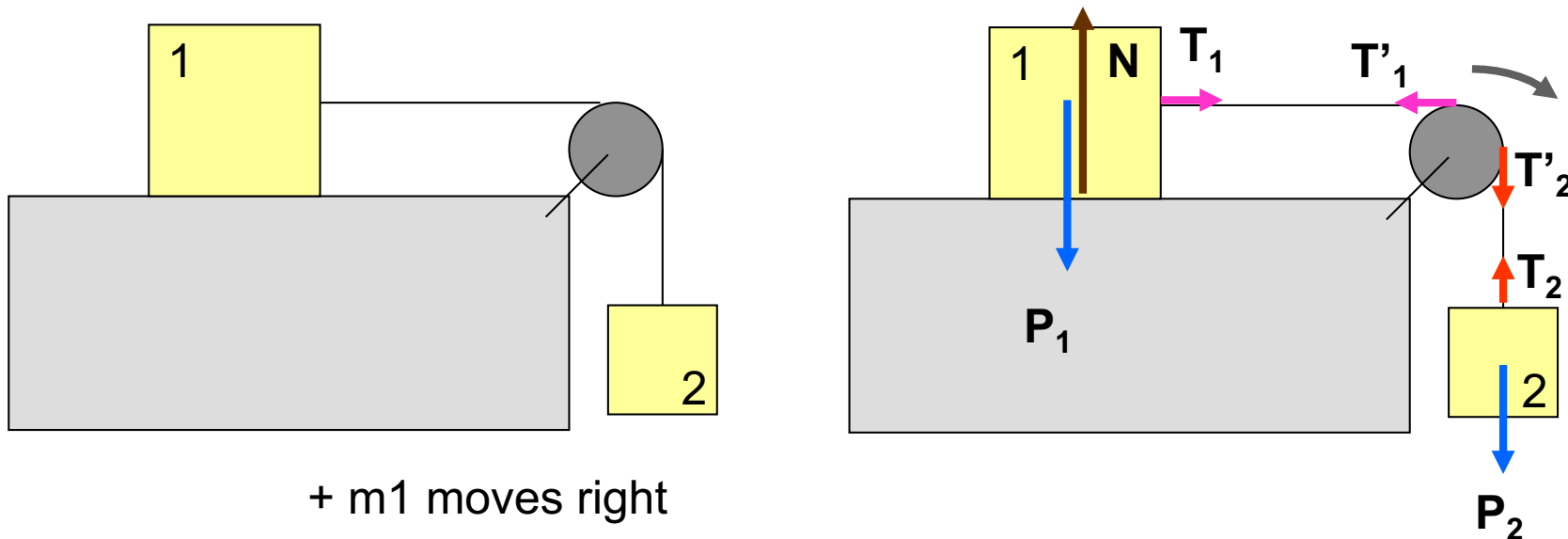
A uniform thin stick of length L and mass M is pivoted at one end. It is held horizontal and released (Figure 9-24). Assume the pivot is frictionless. Find (a) the angular acceleration of the stick immediately after it is released, and (b) the force F_0 exerted on the stick by the pivot at this time.

Example 3

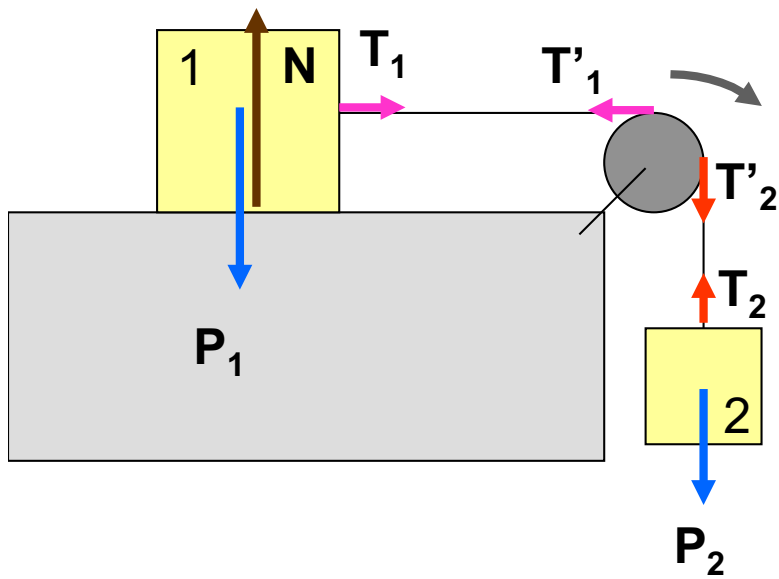
A 4-kg block resting on a frictionless horizontal ledge is attached to a string that passes over a pulley and is attached to a hanging 2-kg block (Figure 9-45). The pulley is a uniform disk of radius 8 cm and mass 0.6 kg. (a) Find the speed of the 2-kg block after it falls from rest a distance of 2.5 m. (b) What is the angular velocity of the pulley at this time?

Example 1

Two blocks are connected by a string that passes over a pulley of radius R and moment of inertia I . The block of mass m_1 slides on a frictionless, horizontal surface; the block of mass m_2 is suspended from the string. Find the acceleration a of the blocks and the tensions T_1 and T_2 assuming that the string does not slip on the pulley.



- + m_1 moves right
- + m_2 moves down
- + Pulley rotates clockwise
- + The magnitude of acceleration : $a_1 = a_2 = a$
- + The magnitude of the tensions: $T_1 = T'_1$, $T_2 = T'_2$



Equation of rotation of pulley

Relationship between a and α

Applying Newton's 2nd law and projecting (1) and (2) on the direction of motion :

$$m_1 \vec{a}_1 = \vec{P}_1 + \vec{N} + \vec{T}_1 \quad (1)$$

$$m_2 \vec{a}_2 = \vec{P}_2 + \vec{T}_2 \quad (2)$$

$$m_1 a_1 = T_1 \quad (3)$$

$$m_2 a_2 = m_2 g - T_2 \quad (4)$$

$$I_{\text{cm}} \alpha = T_2' R - T_1' R \quad (5)$$

$$a = \alpha R \quad (6)$$

$$I_{\text{cm}} \frac{a}{R} = T_2' - T_1' \quad (7)$$

$$(3)+(4)+(7)$$

$$(3)$$

$$(4)$$

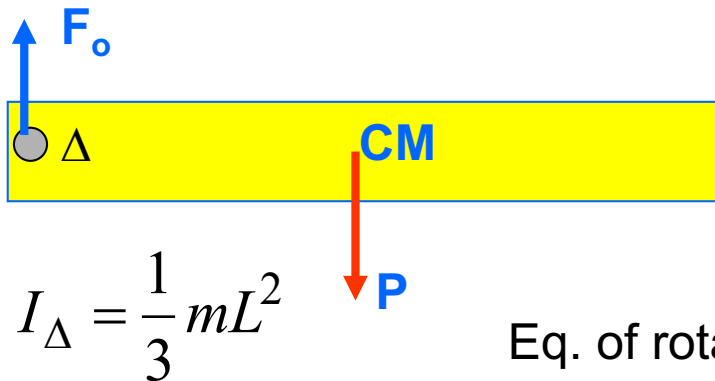
$$\left(m_1 + m_2 + \frac{I_{\text{cm}}}{R^2} \right) a = m_2 g \quad a = \frac{m_2 g}{m_1 + m_2 + \frac{I}{R^2}_{\text{cm}}}$$

$$T_1 = m_1 a_1 = \frac{m_1 m_2 g}{m_1 + m_2 + \frac{I}{R^2}_{\text{cm}}}$$

$$T_2 = m_2 g - m_2 a = \frac{m_1 + \frac{I}{R^2}_{\text{cm}}}{m_1 + m_2 + \frac{I}{R^2}_{\text{cm}}} m_2 g$$

Example 2

A uniform thin stick of length L and mass M is pivoted at one end. It is held horizontal and released (Figure 9-24). Assume the pivot is frictionless. Find (a) the angular acceleration of the stick immediately after it is released, and (b) the force F_o exerted on the stick by the pivot at this time.



Eq. of rotation:

Applying the Newton's 2nd law for the CM gives:

$$ma_{cm} = mg - F_o \quad (1)$$

$$I_{\Delta}\alpha = mg \frac{L}{2} \quad (2)$$

$$\frac{1}{3}mL^2\alpha = mg \frac{L}{2}$$

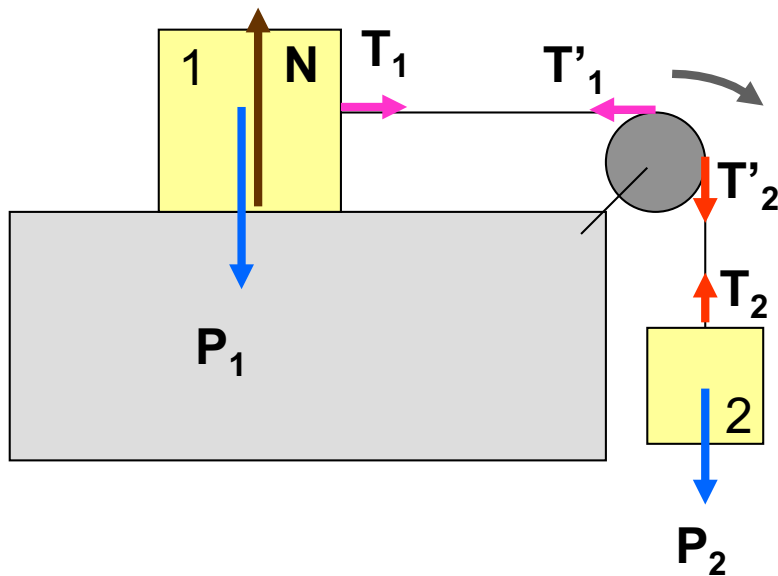
$$\alpha = \frac{3g}{2L} (\text{rad} / \text{s}^2)$$

$$a_{cm} = \alpha \frac{L}{2} = \frac{3}{4}g$$

Remark Just after the stick is released, the pivot exerts an upward force equal to one-fourth the weight of the stick.

$$(1) \quad F_o = mg - ma_{cm} = \frac{1}{4}mg$$

Ex 3 . A 4-kg block resting on a frictionless horizontal ledge is attached to a string that passes over a pulley and is attached to a hanging 2-kg block. The pulley is a uniform disk of radius 8 cm and mass 0.6 kg. (a) Find the speed of the 2-kg block after it falls from rest a distance of 2.5 m. (b) What is the angular velocity of the pulley at this time?



From Ex 9-7
$$a = \frac{m_2 g}{m_1 + m_2 + I / R^2}$$

The motion of m_2 is linear with constant acceleration

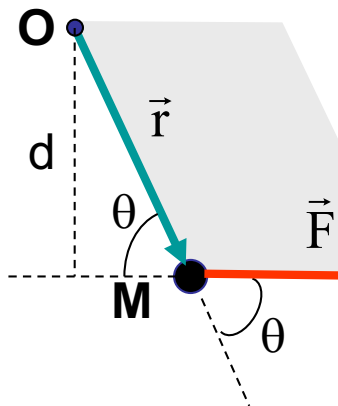
$$v^2 - \underbrace{v_o^2}_{=0} = 2as$$

The speed
$$v = \sqrt{2as}$$

the angular velocity of the pulley
$$\omega = v / R$$

Angular Momentum

- Angular Momentum
- Conservation of Angular Momentum



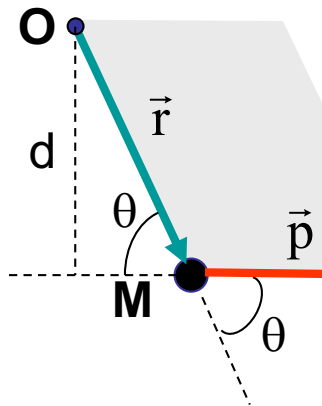
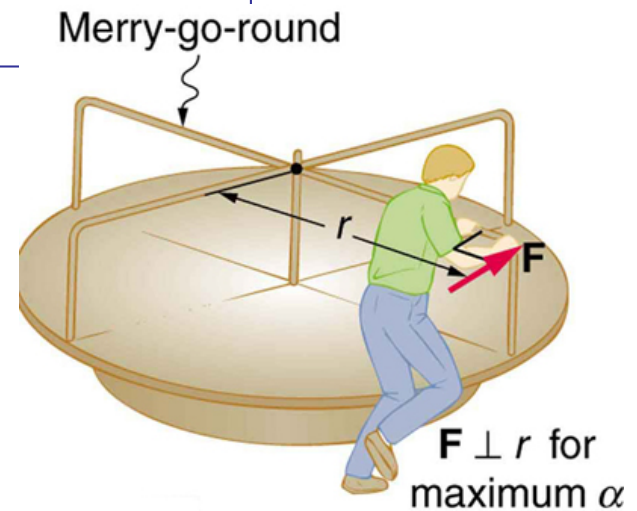
Torque of a force about a fixed point O

$$\vec{\tau}_{\vec{F}/O} = \vec{r} \wedge \vec{F}$$

$$\vec{\tau}_{\vec{F}/O} \perp \text{plane}(\overrightarrow{OM}, \vec{F})$$

The direction of $\vec{\tau}$ is outward

$$\tau_{F/O} = F r \sin \theta = F \cdot d$$



Angular momentum about a fixed point O

$$\vec{L}_{/O} = \vec{r} \wedge \vec{p} = \vec{r} \wedge m\vec{v}$$

$$\vec{L}_{/O} \perp \text{plane}(\overrightarrow{OM}, \vec{p})$$

The direction of $\vec{L}$ is outward

$$L = mvr \sin \theta = mv \cdot d$$

Angular momentum and Torque

Angular momentum about a fixed point O

$$\vec{L}_{/O} = \vec{r} \wedge \vec{p} = \vec{r} \wedge m\vec{v}$$

Differentiate by time:

$$\frac{d\vec{L}_{/O}}{dt} = \frac{d\vec{r}}{dt} \wedge \vec{p} + \vec{r} \wedge \frac{d\vec{p}}{dt} = \underbrace{(\vec{v} \wedge m\vec{v})}_{=0} + \underbrace{\vec{r} \wedge \vec{F}}_{\vec{\tau}}$$

$$\frac{d\vec{L}_{/O}}{dt} = \vec{\tau}_{F/O} \quad \frac{d\vec{L}_{/\Delta}}{dt} = \vec{\tau}_{F/\Delta}$$

The net torque acting on a particle equals the rate of change of the angular momentum of the particle.

For a system of particles

Angular momentum of i^{th} particle about a fixed point O

$$\vec{L}_{i/O} = \vec{r}_i \wedge \vec{p}_i = \vec{r}_i \wedge m\vec{v}_i$$

Angular momentum of the system of particles

$$\vec{L}_{/O} = \sum_i \vec{L}_{i/O} = \sum_i \vec{r}_i \wedge m\vec{v}_i$$

Differentiate by time:

$$\frac{d\vec{L}_{/O}}{dt} = \sum_i \left(\frac{d\vec{r}_i}{dt} \wedge \vec{p}_i \right) + \sum_i \left(\vec{r}_i \wedge \frac{d\vec{p}_i}{dt} \right) = \sum_i \underbrace{(\vec{v}_i \wedge m\vec{v}_i)}_{=0} + \sum_i \vec{r}_i \wedge (\vec{F}_{i,\text{inter}} + \vec{F}_{i,\text{ext}})$$

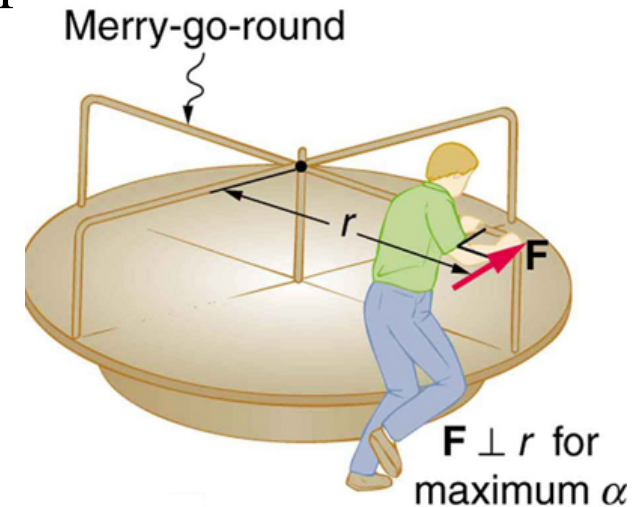
$$\sum_i \vec{r}_i \wedge \vec{F}_{i,\text{inter}} = \vec{0}, \quad \sum_i \vec{r}_i \wedge \vec{F}_{i,\text{ext}} = \sum_i \vec{\tau}_{i,\text{ext}} = \vec{\tau}_{\text{net,ext}}$$

$$\frac{d\vec{L}_{/O}}{dt} = \vec{\tau}_{\text{net,ext}/O}$$

$$\frac{d\vec{L}_{/\Delta}}{dt} = \vec{\tau}_{\text{net,ext}/\Delta}$$

$$\vec{\tau}_{\text{net,ext}/\Delta} = I\vec{\alpha} = I \frac{d\vec{\omega}}{dt}$$

$$\frac{d\vec{L}_{/\Delta}}{dt} = I \frac{d\vec{\omega}}{dt} \Rightarrow \vec{L}_{/\Delta} = I_{\Delta} \vec{\omega}$$

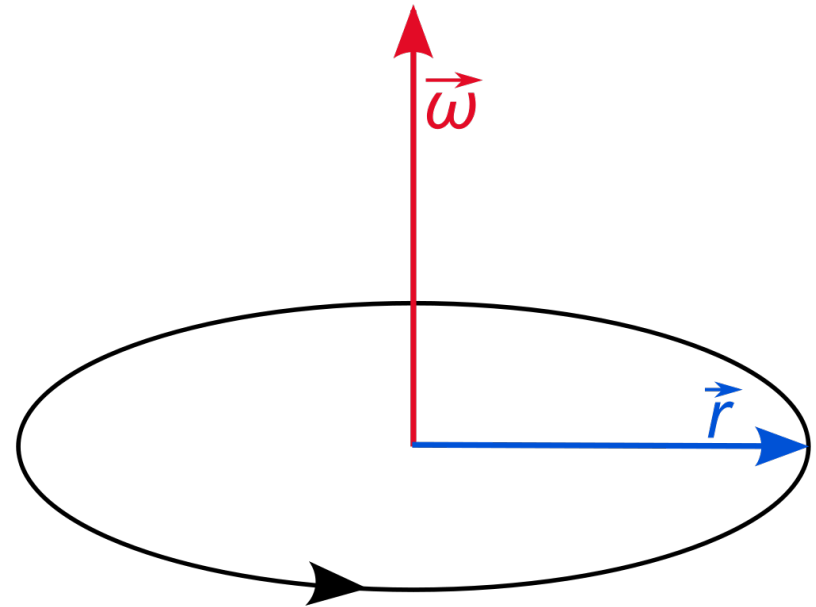


ANGULAR MOMENTUM OF A RIGID OBJECT

$$I_{\Delta} \vec{\alpha} = I_{\Delta} \frac{d\vec{\omega}}{dt} = \vec{\tau}_{\text{net.ext}} / \Delta$$

$$\vec{L}_{\Delta} = I_{\Delta} \vec{\omega}$$

$$\frac{d\vec{L}_{\Delta}}{dt} = \vec{\tau}_{\text{net.ext.}} / \Delta$$



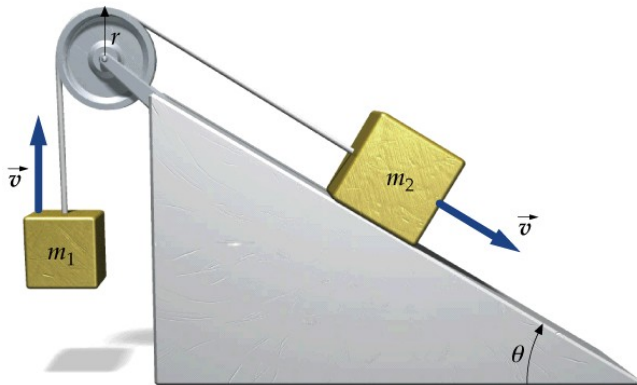
The net external torque acting on a system equals the rate of change of the angular momentum of the system.

$$\frac{d\vec{L}_{/\Delta}}{dt} = \vec{\tau}_{\text{net,ext}/\Delta}$$

$$\vec{L}_{/\Delta} = I_{\Delta} \vec{\omega}$$

Problem

In the figure the incline is frictionless and the string passes through the center of mass of each block. The pulley has a moment of inertia I and a radius r . (a) Find the net torque acting on the system (the two masses, string, and pulley) about the center of the pulley. (b) Write an expression for the total angular momentum of the system about the center of the pulley when the masses are moving with a speed v . (c) Find the acceleration of the masses from your results for parts (a) and (b) by setting the net torque equal to the rate of change of the angular momentum of the system.



$$\text{a) } \tau = m_2 g \sin \theta R - m_1 g R$$

$$\text{b) } L = m_1 v R + m_2 v R + I \omega$$

$$\text{c) } \frac{dL}{dt} = \tau$$

$$(m_1 + m_2 + \frac{I}{R^2}) a R = m_2 g \sin \theta R - m_1 g R$$

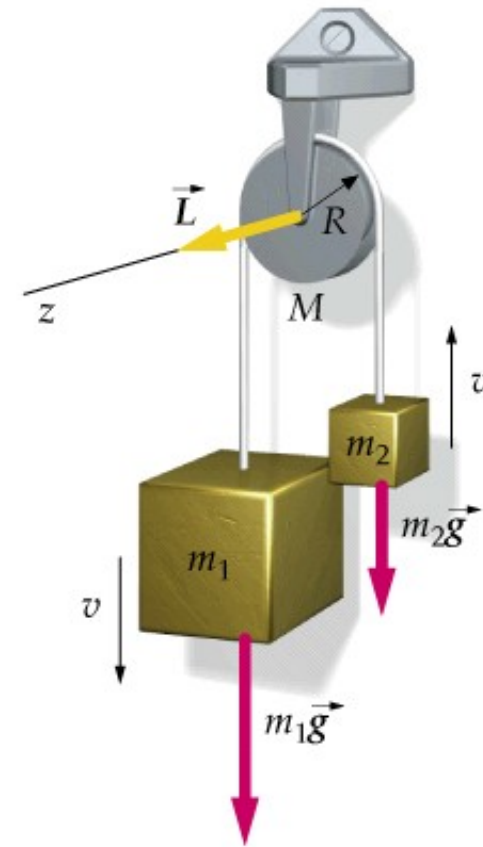
$$a = \frac{m_2 \sin \theta - m_1}{m_1 + m_2 + \frac{I}{R^2}} g$$

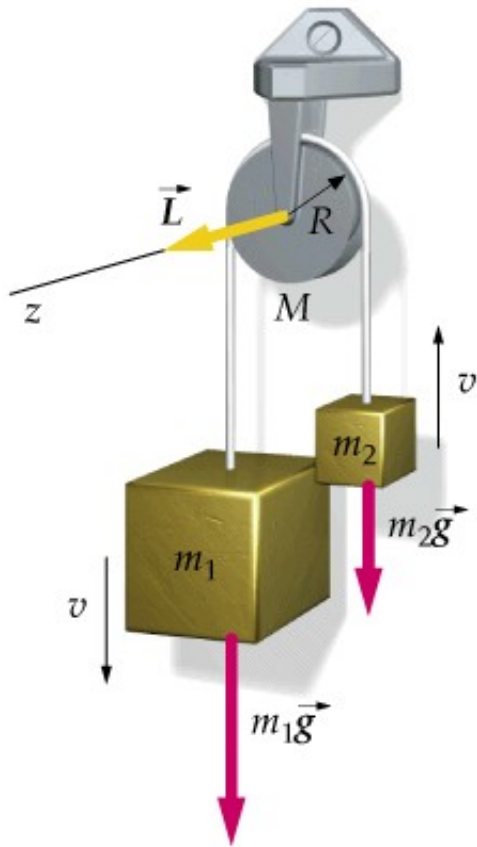
Example 10-2

An Atwood's machine has two blocks of mass m_1 and m_2 ($m_1 > m_2$), connected by a string of negligible mass that passes over a pulley with frictionless bearings. The pulley is a uniform disk of mass M and radius R . The string does not slip on the pulley.

Apply Equation $\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{net,ext}}$

to the system consisting of both blocks plus pulley to find the angular acceleration of the pulley and the acceleration of the blocks.





$m_1 > m_2 \Rightarrow m_1$ moves down,
 m_2 moves up with the speed v
 + pulley rotates counter - clockwise
 with $\vec{\omega}$ outward.

Angular momentum of the system

{Block m_1 , block m_2 , pulley M) about the axis z through the center of pulley

$$L = m_1 v R + m_2 v R + I \omega$$

because that the rope is not sliding on the pulley.

$$v = \omega R \Rightarrow a = \alpha R$$

$$\frac{dL}{dt} = m_1 a R + m_2 a R + I \alpha = \left(m_1 + m_2 + \frac{I}{R^2} \right) a R$$

Net torque of external forces :

$$\tau_{\text{net}} = m_1 g R - m_2 g R$$

$$\text{Applying } \frac{dL}{dt} = \tau_{\text{net, ext}}$$

$$\left(m_1 + m_2 + \frac{I}{R^2} \right) a R = m_1 g R - m_2 g R$$

$$a = \frac{m_1 - m_2}{m_1 + m_2 + \frac{I}{R^2}} g$$

Conservation of Angular Momentum

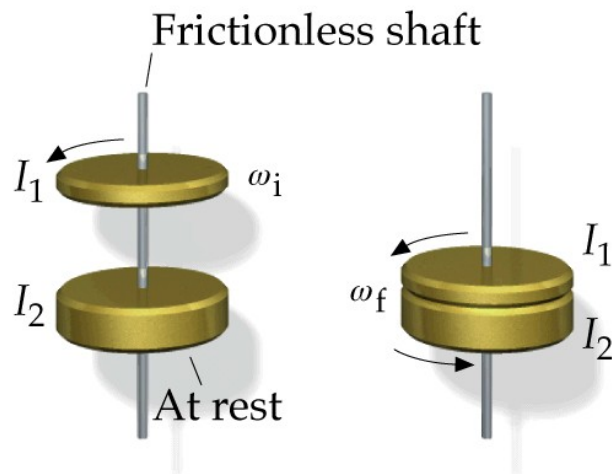
$$\frac{d\vec{L}}{dt} = \vec{\tau}_{\text{net,ext}}$$

$$\text{if } \vec{\tau}_{\text{net,ext}} = 0 \quad \vec{L} = \text{const}$$

If the net external torque acting on a system is zero, the total angular momentum of the system is constant.

Example

A disk is rotating with an initial angular speed ω_i about a frictionless shaft through its symmetry axis as shown. Its moment of inertia about this axis is I_1 . It drops onto another disk of moment of inertia I_2 that is initially at rest on the same shaft. Because of surface friction, the two disks eventually attain a common angular speed ω_f . Find ω_f



External force : weight m_1g parallel to the rotation axis

$\Rightarrow$ torque $\vec{\tau}_{\text{net}} = 0 \Rightarrow \vec{L} = \text{const}$

initial Angular momentum = final angular momentum

$$L_i = L_f$$

$$I_1 \vec{\omega}_i = (I_1 + I_2) \vec{\omega}_f \Rightarrow \vec{\omega}_f = \frac{I_1}{I_1 + I_2} \vec{\omega}_i$$

$$\text{Initial KE : } T_i = \frac{1}{2} I_1 \omega_i^2 = \frac{L^2}{2I_1}$$

$$\text{Final KE : } T_f = \frac{1}{2} (I_1 + I_2) \omega_f^2 = \frac{L^2}{2(I_1 + I_2)}$$

The change in KE :

$$\Delta T = T_f - T_i = \frac{L^2}{2} \left(\frac{1}{I_1 + I_2} - \frac{1}{I_1} \right) = -\frac{L^2}{2} \frac{I_2}{(I_1 + I_2)I_1} < 0$$

Example

A merry-go-round of radius 2 m and moment of inertia 500 kg.m^2 is rotating about a frictionless pivot, making one revolution every 5 s. A child of mass 25 kg originally standing at the center walks out to the rim.

Find the new angular speed of the merry-go-round.

External force is the gravitational force $m\vec{g}$, parallel to the rotation axis. The torque is zero.

Applying the law of Conservation of Angular momentum $L_i = L_f$

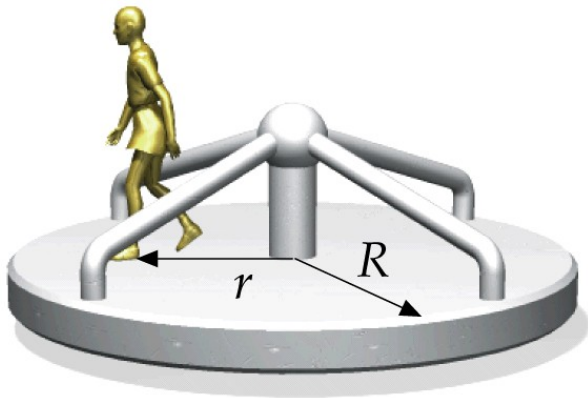
$$(I_{\text{merry}} + I_{\text{child},i})\omega_i = (I_{\text{merry}} + I_{\text{child},f})\omega_f$$

$$\omega_f = \frac{I_{\text{merry}} + I_{\text{child},i}}{I_{\text{merry}} + I_{\text{child},f}} \omega_i$$

$$I_{\text{child},i} = 0 \quad I_{\text{child},f} = mR^2$$

$$\omega_f = \frac{I_{\text{merry}}}{I_{\text{merry}} + mR^2} \omega_i$$

$$\omega_f = \frac{500}{500 + 25 \times 2^2} \frac{2\pi}{5} = 1.05 \text{ rad/s}$$



3.3. Rolling without sliding

https://www.youtube.com/watch?v=r_yqJ2HXoC0

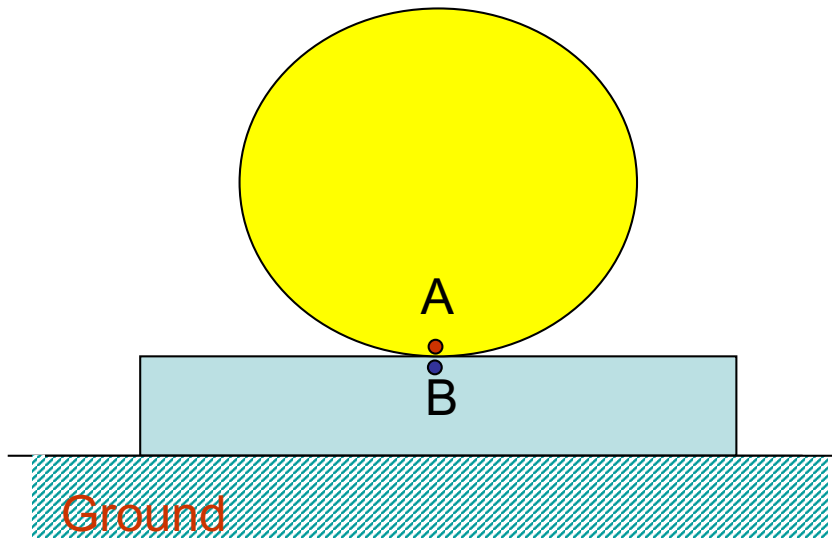
What does it mean “Without sliding”?

The sphere is not sliding relative to the bar
if the velocities of the two adjacent points A and B are the same.
where A is on the sphere and B is on the bar

Without sliding : $\vec{v}_{A/\text{ground}} = \vec{v}_{B/\text{ground}}$ or $\vec{v}_A = \vec{v}_B$

The velocity of A relative to B is :

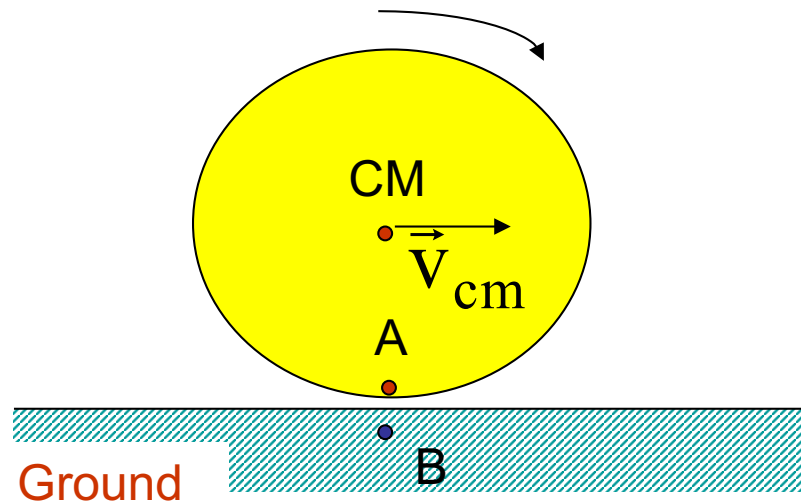
$$\vec{v}_{A/B} = \vec{v}_{A/\text{ground}} - \vec{v}_{B/\text{ground}} = 0$$



If there is 'sliding':

$$\vec{v}_A \neq \vec{v}_B \text{ and } \vec{v}_{A/B} \neq 0$$

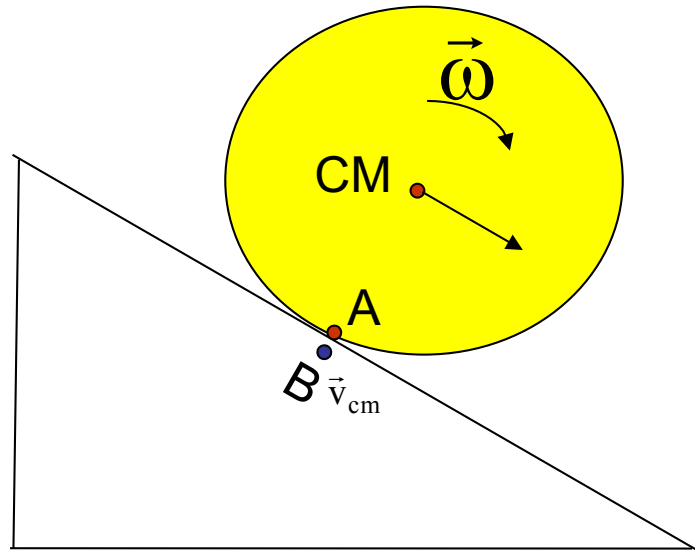
Rotation about a moving axis = Rolling



A cylinder is rolling on the ground.

Two types:

- Rolling with sliding
- Rolling without sliding



https://www.youtube.com/watch?v=PWvIYU_Z8z8

Rolling

=

Translation
of a point of the object

+

Rotation
about the axis
through this point

Rolling

=

Translation
of the center of mass

+

Rotation
about the axis
through CM

Rolling

=

Translation
of point A
(center
of instantaneous velocity)

+

Rotation
about the axis
through A

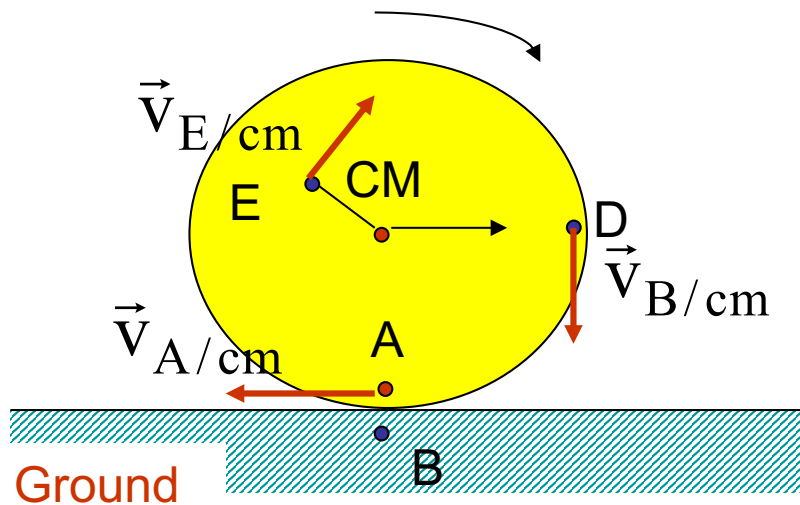
Velocity in rolling motion

Velocity of a point M of the rolling object.

$$\vec{V}_{M/\text{ground}} = \vec{V}_{M/\text{cm}} + \vec{V}_{\text{cm}/\text{ground}}$$

Velocity of a point M in the rotation about the axis through CM

$$\vec{V}_{M/\text{cm}} = \vec{\omega} \wedge \overrightarrow{\text{CM}}$$

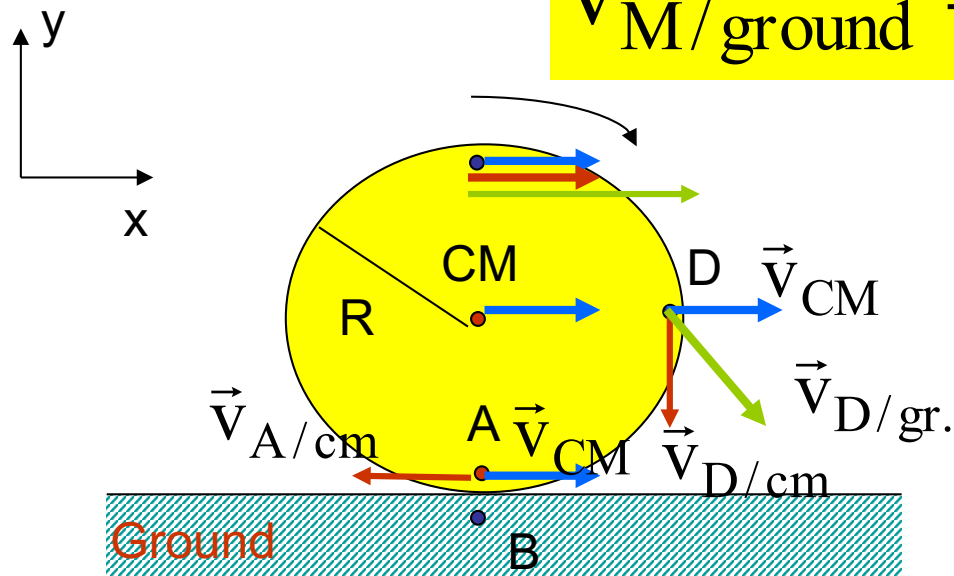


$$|\vec{V}_{A/\text{cm}}| = \omega R$$

$$|\vec{V}_{D/\text{cm}}| = \omega R$$

$$|\vec{V}_{E/\text{cm}}| = \omega \frac{R}{2}$$

$$\vec{V}_{M/\text{ground}} = \vec{V}_{M/\text{cm}} + \vec{V}_{\text{cm}/\text{ground}}$$



Rolling without sliding,

$$\vec{V}_A = 0$$

because the ground is not moving, $v_B = 0$

$$\vec{V}_A = \vec{V}_{A/\text{cm}} + \vec{V}_{\text{cm}} = 0 \quad (1)$$

Projecting (1) on the x axis :

$$-\omega R + v_{\text{cm}} = 0$$



$$\begin{aligned} \omega R &= v_{\text{cm}} \\ \alpha R &= a_{\text{cm}} \end{aligned}$$

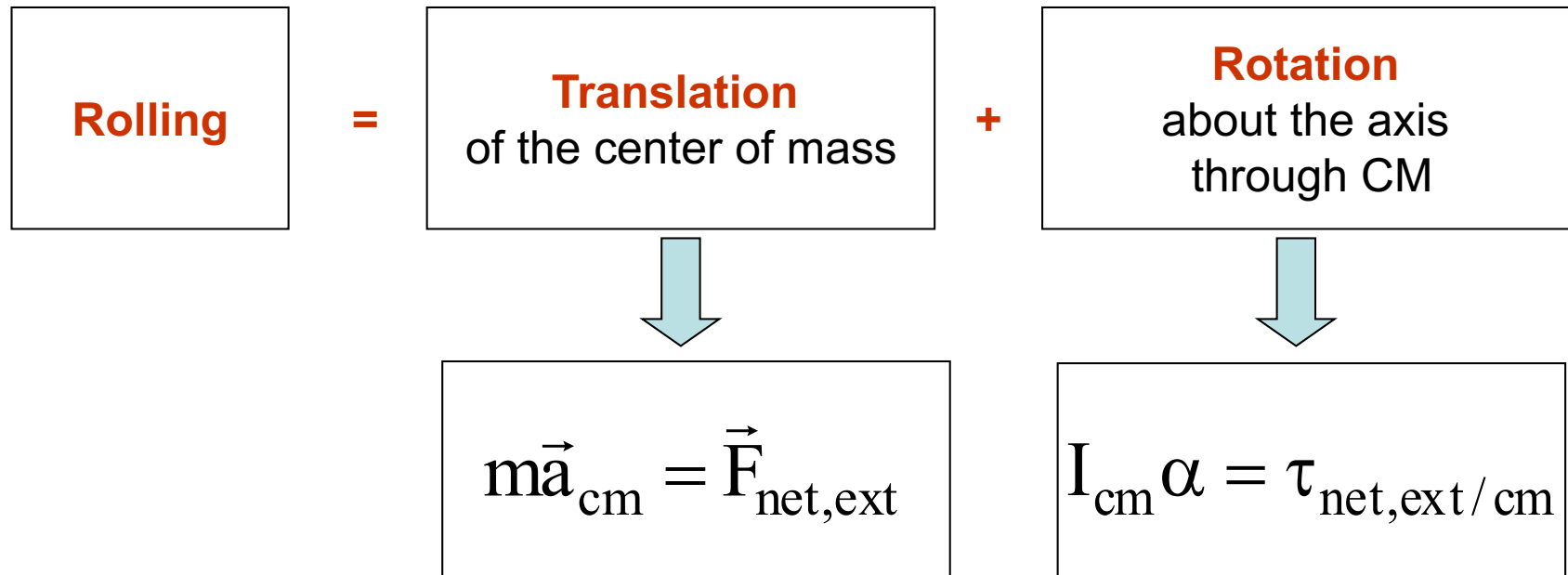
ω : angular velocity in rotational motion

α : angular acceleration

$V_{\text{cm}}, a_{\text{cm}}$: speed and acceleration of the CM

How to Solve Problems 'Rolling without Sliding'

Method 1



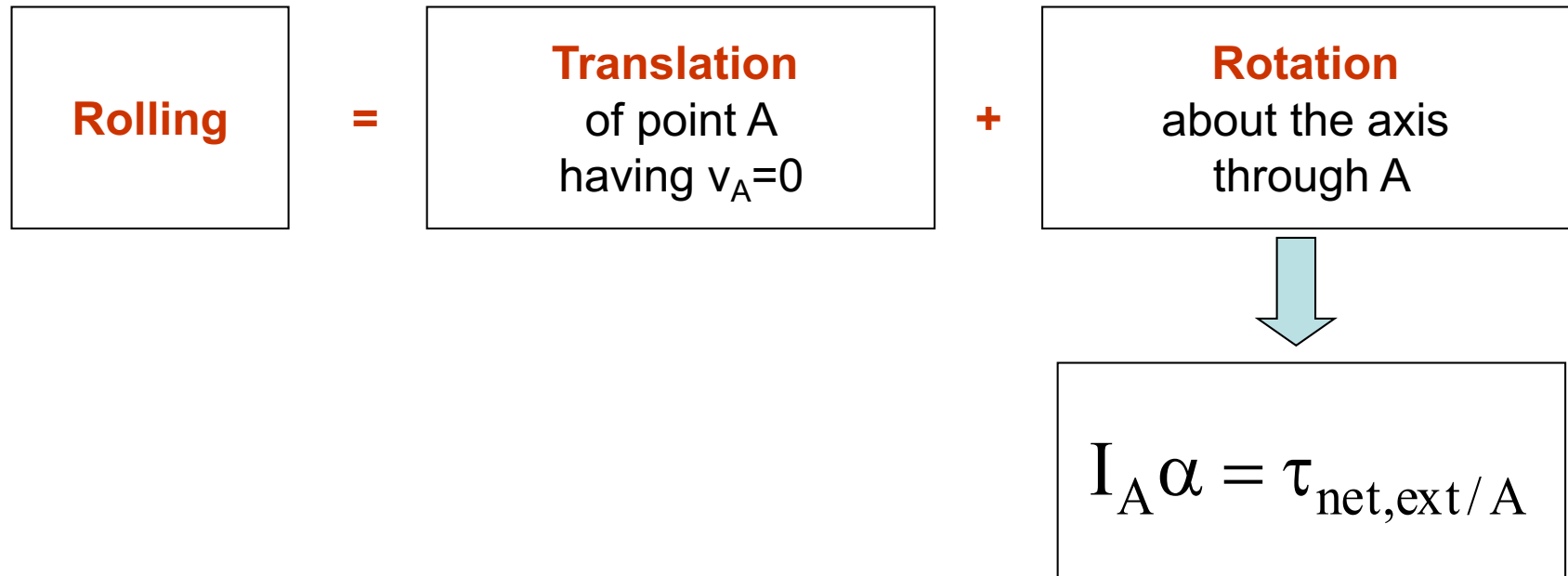
Step 1. Write Eq. of Motion of the CM

Step 2. Write Eq. of Rotation about the axis through CM

Step 3. Find relation between a_{cm} and α

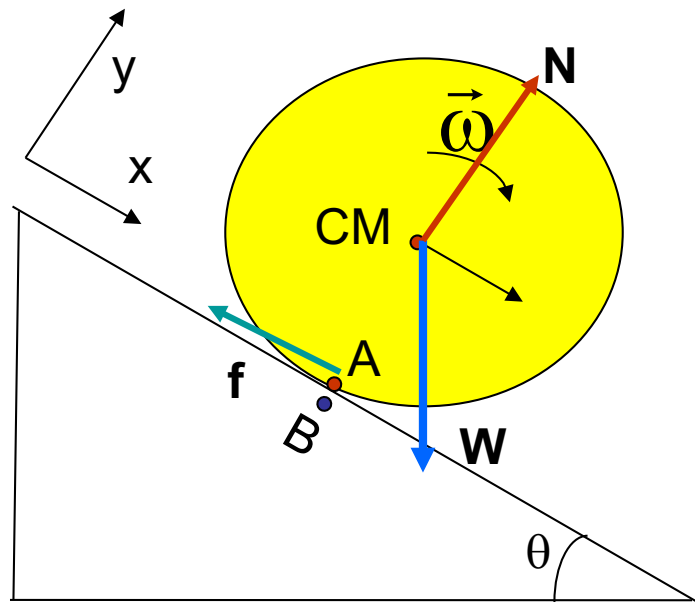
How to Solve Problems 'Rolling without Sliding'

Method 2



Step 1. Write Eq. of Rotation about the axis through A $\Rightarrow \alpha$

Step 2. Find relation between a_{cm} and $\alpha \Rightarrow a_{\text{cm}}$



Example

Method 1.

A cylinder of mass m , radius R rolls without sliding on an inclined plane.

Find

- acceleration of CM
- angular acceleration
- friction force
- condition for having 'rolling without sliding'

Step1. Eq. of motion of CM : $m\vec{a}_{cm} = m\vec{g} + \vec{N} + \vec{f}_{friction}$ (1)

Projecting on x and y axis

(1) / x : $ma_{cm} = mg \sin \theta - f_{friction}$ (2)

(1) / y : $0 = -mg \cos \theta + N$ (3) $\Rightarrow N = mg \cos \theta$ (3')

Step2. Eq. of Rotation about the axis through CM :

$I_{cm} \alpha = f_{friction} R$ (4)

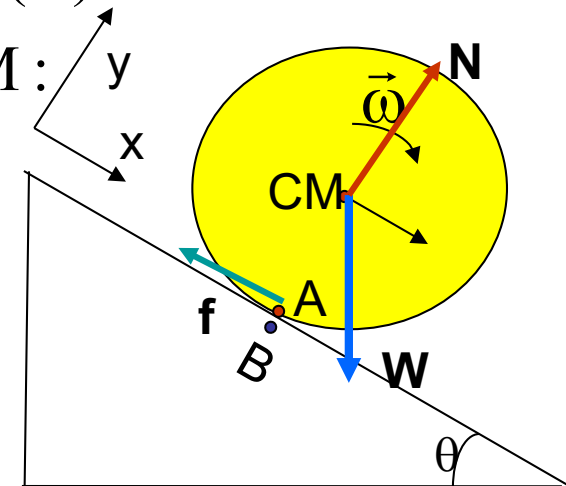
Step3. Relation between a_{cm} and α :

$a_{cm} = \alpha R$ (5)

From (4), (5) : $\frac{I_{cm}}{R^2} a_{cm} = f_{friction}$ (6)

(2) + (6) : $(m + \frac{I_{cm}}{R^2}) a_{cm} = mg \sin \theta \Rightarrow a_{cm} = \frac{mg \sin \theta}{m + \frac{I_{cm}}{R^2}}$

from (2) : $f_{friction} = mg \sin \theta - ma_{cm}$

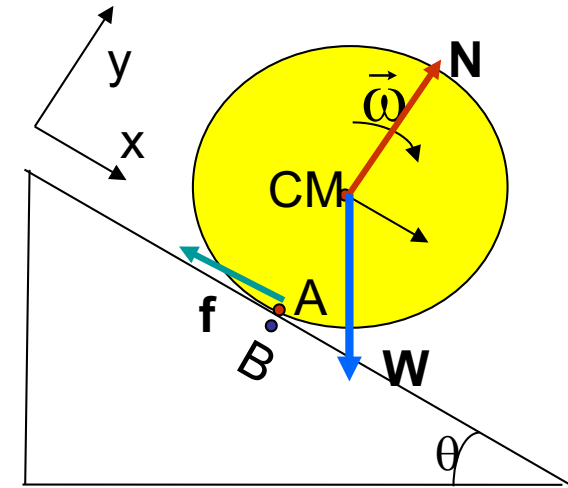


$$a_{\text{cm}} = \frac{mg \sin \theta}{m + \frac{I_{\text{cm}}}{R^2}}$$

$$\alpha = a_{\text{cm}} / R$$

$$N = mg \cos \theta$$

$$f_{\text{friction}} = mg \sin \theta - ma_{\text{cm}}$$



a) solid cylinder: $I_{\text{cm}} = \frac{1}{2} mR^2$

$$a_{\text{cm}} = \frac{2}{3} g \sin \theta; \quad f_{\text{friction}} = \frac{1}{3} mg \sin \theta$$

b) solid sphere: $I_{\text{cm}} = \frac{2}{5} mR^2$

$$a_{\text{cm}} = \frac{5}{7} g \sin \theta; \quad f_{\text{friction}} = \frac{2}{7} mg \sin \theta$$

b) Hoop: $I_{\text{cm}} = mR^2$

$$a_{\text{cm}} = \frac{1}{2} g \sin \theta; \quad f_{\text{friction}} = \frac{1}{2} mg \sin \theta$$

• condition for having
'rolling without sliding'

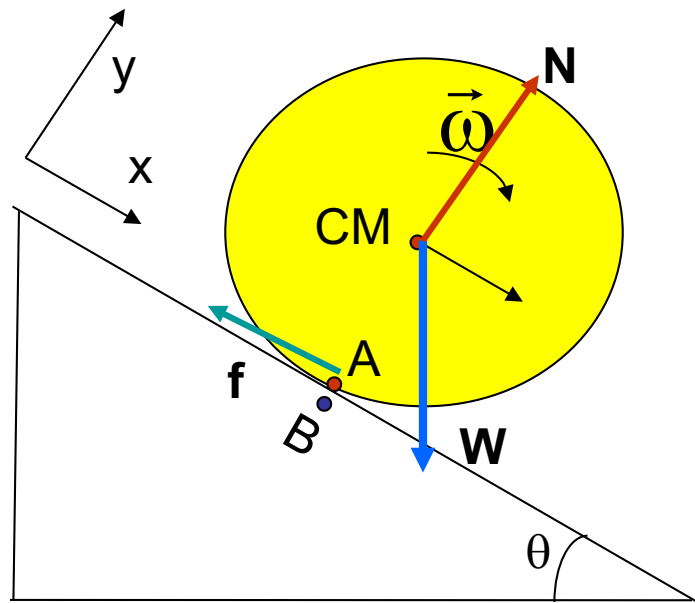
$$f_{\text{fric}} \leq f_{\text{smax}}$$

$$f_{\text{fric}} \leq \mu_s N$$

Cylinder

$$\frac{1}{3} mg \sin \theta \leq \mu_s mg \cos \theta$$

$$\tan \theta \leq 3\mu_s$$



Example

Method 2.

A cylinder of mass m , radius R rolls without sliding on an inclined plane.

Find

- acceleration of CM
- angular acceleration
- friction force
- condition for having 'rolling without sliding'

Eq. of Rotation about the axis through A:

$$I_A \alpha = mgR \sin \theta \quad \Rightarrow \quad \alpha = \frac{mgR \sin \theta}{I_A}$$

Parallel Axis Theory: $I_A = I_{cm} + mR^2 = \frac{3}{2}mR^2$

$$\Rightarrow \quad \alpha = \frac{2}{3R} g \sin \theta$$

$$\Rightarrow \quad a_{cm} = \alpha R = \frac{2}{3} g \sin \theta$$

$$\alpha = \frac{2}{3R} g \sin \theta$$

$$a_{cm} = \frac{2}{3} g \sin \theta$$

Find $f_{fric.}$ from

$$a) \quad ma_{cm} = mg \sin \theta - f_{friction}$$

$$\text{or } b) \quad I_{cm} \alpha = f_{friction} \cdot R$$

Rolling without and with sliding

For rolling without sliding :

$$f_{\text{friction}} = f_{\text{static}} \leq f_{\text{s. max}}$$

$$v_{\text{cm}} = \omega R$$

$$a_{\text{cm}} = \alpha R$$

Work of static friction force is zero

For rolling with sliding :

$$f_{\text{friction}} = \mu_k \cdot N$$

$$v_{\text{cm}} \neq \omega R$$

$$a_{\text{cm}} \neq \alpha R$$

$$W_{\text{fric}} = -\mu_k N \cdot s$$

Kinetic Energy in Rolling

$$K = \underbrace{\frac{1}{2}mv_{\text{cm}}^2}_{\text{K.E of Translational motion of CM}} + \underbrace{\frac{1}{2}I_{\text{cm}}\omega^2}_{\text{K.E of Rotational Motion about the axis through CM}}$$

1) A cylinder rolls down a plane inclined at $\theta = 50^\circ$.
What is the minimum value of the coefficient of static friction for which the cylinder will roll without slipping? (Answer 0.40)

2) For a hoop rolling down an incline,
(a) what is the force of friction,
(b) what is the maximum value of $\tan\theta$ for which the hoop will roll without slipping?
(Answers (a) $f = \frac{1}{2}mg\sin\theta$, (b) $\tan\theta = 2\mu_s$)

Example

Investigate the motion of a hollow cylinder that is rolled downwards as shown. We know that the mass of the cylinder is m and the radius of the cylinder is R . The string is unstretchable and massless.

